

**Volume Effect on Return and Volatility Spillover in  
Dually Listed Securities**

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## **Abstract**

We empirically investigate the volume impact on serial correlations in return and volatility of 170 ADRs listed in the NYSE market and their underlying securities from 29 developed and emerging markets in the world and those of their underlying home shares. Using a VAR setup, we forecast volume of cross listed securities based on Halling et al. (2013) and hence denote the residuals as unanticipated volume. We find this unanticipated volume or volume surprise has an asymmetric effect on return and volatility spillover, serial auto and cross correlations between ADRs and their underlying home shares. Specifically, we find the effect of volume on serial cross correlations for both return and volatility spillover are more pronounced in home shares and diametrically opposite (continuation vs. reversal) to those in ADRs.

## **1. Introduction**

Using a panel of daily data on 170 ADRs listed on the New York Stock Exchange (NYSE later) and their underlying securities from 29 developed and emerging markets around the world over a period of 8-23 years, we empirically determine the impact of volume on the return and volatility dynamics of cross listed securities. Specifically, we address the following questions: How do trading volumes of cross listed securities commove over time? How does volume surprise or unanticipated volume adjusted for such comovements in volumes between pairs of cross listings impact return and volatility spillover, serial correlations of and between ADRs and their underlying home securities?<sup>1</sup> Finally, are there common market/country and security specific determinants of volume surprise induced return and volatility spillover?

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<sup>1</sup> We use the term spillover rather broadly to indicate serial autocorrelations for a security and also serial cross correlations between matching pairs of securities. Note that while serial correlations in returns are mentioned in the literature as spillover, serial autocorrelations in volatility are mostly referred to as persistence and only serial cross correlations as volatility spillover in cross market analysis. We also use the terms volume surprise and unanticipated volume interchangeably throughout the paper. In this paper, our primary interest is to analyze the volume impact on serial cross correlations in returns and volatility persistence.

Two existing strands of empirical research on return and volatility spillover provide the foundation of our investigation. First, Conrad et al. (1994) observe in a sample of NASDAQ stocks that the volume impact on first order autocorrelation in return or return spillover in individual stocks is asymmetric such that high (low) trading volume stocks experience return reversals (continuation). Llorente et al. (2002) broaden the sample to include all US stocks listed on the NYSE and the AMEX and further explain the evidence on return spillover as due to high (low) level of information (hedging or allocation) demands as implied by Campbell et al. (1993) and observed in small (large) cap stocks.<sup>2</sup> Gagnon and Karolyi (2009) extend this literature to include cross listed securities and document that firms with low (high) levels of information asymmetry witness return reversals (weaker or continuation) in one market following high volume in the alternate market; in addition, they report that return spillover, be it continuation or reversal, is more pronounced when it originates from the home market rather than the host (US) market.

Second, while evidence of serial autocorrelations in volatility or volatility clustering abounds in the finance literature, limited evidence exists regarding cross security volatility spillover i.e., volatility transmission from ADR (host) to its underlying (home) and reverse.<sup>3</sup> Chan et al. (1996) compare information transmission

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<sup>2</sup> Recently, following up on Gervais et al. (2001), Heston et al. (2010) find evidence of high volume premium, that is, returns are higher for actively traded securities and during periods of high trading, for example, during the opening hours of trading. Kaniel (2012) find evidence of volume premium in 41 countries in the world. Note that in this paper, our interest is not on volume premium but the impact of volume on return spillover.

<sup>3</sup> Refer to the literature surveys on GARCH regarding volatility clustering and persistence. Much of the empirical literature on intercountry return and volatility transmission is based mostly on country indexes; recently, studies report on volatility transmission on country indexes denoted by respective ETFs. For a literature review, refer to Gagnon and Karolyi (2006). Diebold and Yilmaz (2009) and Bekaert et al. (2009, 2014) study volatility transmission across markets to test time varying financial market integration and spread of contagion across markets.

between Japanese and European ADRs while Hauser et al. (1998) investigate the information flow between NYSE and Tel Aviv exchange for five dually listed stocks. More recently, Yang (2007) documents cross country volatility spillover for Japanese ADRs and Poshakwale and Aquino (2008) report on volatility transmission for a global basket of 70 ADRs.

Further, with respect to volume impact on time varying serial correlations in volatility of domestic US stocks, Abramov et al. (2006) find empirical support for the implications in Wang (1993, 1994) and Campbell et al. (1993) that volatility must decrease (increase) with informational (hedging or allocation motivated liquidity) trading.<sup>4</sup> Nevertheless, there is no published research to date on how volume or volume surprise affects volatility spillover, serial auto and cross correlations in volatility in cross listed securities in home and host markets.

We observe that the central issue in this literature is how to skim the unexpected information content from trading volume since it's the unanticipated component of volume that affects asset prices and further to use a consistent methodological approach to measure its effect on asset return and volatility. The existing literature on the impact of unanticipated volume on return and volatility spillover, particularly as it relates to cross listed securities fails to consider volume comovements between pairs of cross listed securities for the purpose of computing expected volume for a security, which casts doubt on the results of volume impact on return spillover in Gagnon and Karolyi (2009), who deseasonalize volume for cross listed securities with a 50 days MA filter

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<sup>4</sup> Note that Lo and Mackinlay (1990) and Sentana and Wadhwani (1992) propose a separate market microstructure based asynchronous vs. feedback trading to justify volatility changes in equity markets.

for the purpose of determining unanticipated excess (+/-) volume. Further, since asset pricing models predict both return and variance, this information content in trading volume also likely affects volatility spillover in cross listed shares. Nevertheless, there exists no empirical evidence (that we are aware of) with regards to how volume surprise may affect volatility spillover in and between cross listed securities. In addition, it is also unknown if there are common determinants of the volume induced return and volatility spillover for a diverse sample of cross listed securities.

Empirically, we address these concerns as follows. First, we follow Halling et al. (2013), who test the predictions of theoretical models of multi market trading volume by Chowdhury and Nanda (1991), Pagano (1989), and Menkveld (2008) and run a bivariate VAR model to estimate expected trading volumes of each pair of ADR and its underlying home security, in which the prediction errors denote volume surprises. Second, we estimate the auto and cross correlation coefficients along with two interaction variables denoting volume induced returns, parameters of another bivariate VAR model for ADR and home security returns and revisit Gagnon and Karolyi (2009) to test the implications of Campbell et al. (1993), Wang (1994), and Blume et al. (1994). Third, we extend Abramov et al. (2006) and test the implications of Wang (1993, 1994) in terms of volume impact on volatility spillover for a sample of cross listed securities. We compute squared errors of returns to denote volatility and estimate the parameters of a bivariate VAR model for volatility in which not only lagged variance but lagged cross covariance, and volume surprises induced volatility enter into each security's variance function. Finally, in a system of equations setup, we determine the effect of

security specific and market/country related exogenous variables, for example, developed vs. emerging markets, market capitalization, legal origin, and geography on the spillover parameter estimates.

Our empirical results find the following. First, there is significant evidence of short term memory and volume comovements, albeit asymmetric, across a majority of ADRs but all of their corresponding home shares. Second, first order serial return autocorrelations are polar opposites- negative for ADRs and positive for their underlying home shares, while return cross correlations are uniformly positive for both ADRs and the corresponding home shares. The mean volume induced autocorrelations are generally positive for both ADRs and their underlying home shares indicating continuation, while the mean volume induced cross correlations are positive implying continuation for ADRs but negative denoting reversal for their underlying home shares. Third, tests of volatility spillover indicate significant auto and cross correlation effects in both ADR and their underlying security volatility. However, while volume surprise do not appear to have any bearing on volatility persistence of either ADRs or their underlying home securities, volume surprise show significant reversals in volatility for both ADRs and their underlying home shares. Fourth, we find multidimensional asymmetry in terms of the impact of country/market characteristics on return and volatility spillover, ADRs and their home shares.

Our research contributes to the literature in multiple ways. First, we decompose volume into expected volume and volume surprise based on Halling et al. (2013) multiple market trading model, in which the respective market and cross security

trading volumes are among the determinants of a security's trading volume. Hence, we revisit the evidence in Gagnon and Karolyi (2009) regarding unanticipated volume effect on return spillover from a comprehensive set of dually listed securities from multiple developed and emerging markets. Second, in our knowledge, this is the first study of its kind that examines unanticipated volume effect on volatility spillover for a comprehensive set of dually listed securities from multiple developed and emerging markets. Third, through a robust regression methodology, we determine common security and market/country specific factors which explain variations in the volume induced auto and cross correlation coefficients in return and volatility. Results from our study show short term predictability in return and volatility and hence refute market integration, which in turn has direct practical ramifications for international portfolio management, specifically in asset allocation and risk management.

## **2. Literature Review**

We document the effect of unanticipated trading volume or volume surprise on return and volatility spillover, serial auto and cross correlations for cross listed securities in home and host (NYSE) markets. This research question overlaps diverse areas of the finance literature and hence for the purpose of exposition, we break down the literature as follows: a) Volume surprise model for cross listed securities b) Return and volatility spillover and further unanticipated volume effect on such spillover c) Security and country/market specific determinants of return and volatility spillovers. We provide a brief survey of the existing literature often overlapping two or more topics mentioned

above.

Regarding trading volume modeling for individual securities, Lo and Wang (2010) provide a summary of trading volume definitions used in the literature. Trading volume of single listed shares may contain a unit root, a deterministic or stochastic trend, a long memory component, or even a combination depending on a variety of factors including estimation intervals and sample period (Lo and Wang [2000], Fleming and Kirby [2009], Chae [2005]). Brock and Kleidon (1992), Huberman (), and Brown et al. (2009) find single market trading volume denote multiple motives including information, inventory, and momentum.

For cross listed shares, Pagano (1989), Chowdhury and Nanda (1991), and Menkveld (2008) propose theoretical models of multimarket trading volume with discretionary traders and conclude that any optimal trading rule includes the possibility that all or a substantial portion of trading is concentrated in only one market. Empirically, Halling et al. (2013) find continued dual market trading for cross listed shares and that while there is significant correlation between trading volumes across multiple trading venues, the breakdown of trading volume among those alternative venues is not consistent among firms and markets and further two time varying factors, legal enforcement regime (market/country specific) and institutional ownership (security specific) impact the volume of discretionary trading in each market.

Regarding trading volume impact on autocorrelation and/or cross correlation between returns on domestic securities, Conrad et al. (1994) explore the nexus between trading activity and short horizon (weekly) return comovements for NASDAQ listed



securities and report that high (low) activity stocks experience return reversals (continuation of positive correlation); Llorente et al. (2002) deduce empirical implications from Campbell et al. (1993) and Wang (1994) and explain the return correlation continuation (reversals) for low (high) activity stocks as due to higher (lower) level of information asymmetry and informed trading in small cap stocks, which also exhibit high bid ask spread.

Gagnon and Karolyi (2009) extend the dynamic volume return literature by adding cross listed securities in the mix. In a comprehensive study of 556 foreign securities listed in the US and their underlying home securities, they find that firms with low (high) levels of information asymmetry witness return reversals (weaker or continuation) in one market following high volume in the alternate market; in addition, they report that return spillover, be it continuation or reversal, is more pronounced when it originates from the home market rather than the host (US) market. Empirically, Gagnon and Karolyi (2009) use a two-step process. First, they estimate the autocorrelation and cross correlation parameters along with their interactions with security specific trading volumes from a time series regression model for each security pair over each quarter; and then in a pooled cross sectional regression model in which the volume induced spillover parameters are the dependent variables, they estimate the coefficients associated with several information asymmetry measures as independent variables. He and Yang (2012) compare daytime and overnight return spillover between US and China for a sample of Chinese ADRs.

In terms of volume volatility relation, multiple empirical issues have emerged

including those related to cross listed securities.<sup>5</sup> First, although a long list of studies have investigated whether for broad markets volume and volatility are contemporaneously correlated (MDH) or alternately there exists a bi-directional lead lag relation (SIAH) between the two, only a few studies report on similar volume volatility nexus for individual securities. Bollerslev and Jubinski (1999) find volume and volatility of US stocks are fractionally integrated and share a common long memory parameter; however, Fleming and Kirby (2007) report that 20% of their sample of 20 middle market securities do not have identical long memory in volume and volatility and Anè and Ureche-Rangau (2008) find that volume and volatility share only short term comovement but not any long term memory. On the other hand, Darrat et al. (2003) report one-way Granger causality from trading volume to return volatility for 24 out of 30 DJIA stocks. Li and Wu (2006) reports non-linear correlation between volume and volatility of a sample of US stocks.

A second question even more pertinent to this research is how volume changes the serial auto and cross correlation coefficients associated with volatility of cross listed securities. This question nests three separate queries- How volatility transmits across

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<sup>5</sup> Karpoff (1987) surveys the literature on the relation between changes in trading volume and absolute price changes. Two contrasting hypotheses, the Mixture of Distribution (MDH) and the Sequential Information Arrival (SIAH) hypotheses exist to explain the volume volatility relations. Harris (1986), Smirlock and Starks (1986), Lamoureux and Lastrapes (1990), Kim and Kon (1994), Anderson (1996), and Gallo and Pacini (2000) find evidence in support of a contemporaneous volume volatility relation in the US stock market, as predicted by the MDH. Chen and Tse (1993), Omran and McKenzie (2000), Pyun et al. (2000), and Bohl and Henke (2003) report similar evidence from the Japanese, UK, Spanish, Korean, and Polish stock markets respectively. In contrast, the SIAH (Copeland 1976) predicts bidirectional causality between volume and volatility. Saatcioglu and Starks (1998) find positive correlation between changes in trading volume and price changes but mixed evidence on the direction of causality in Latin American markets. Campbell et al. (1993), Hiemstra and Jones (1995), and Brooks (1998) report bidirectional Granger causality between daily stock returns and changes in trading volume. Gallant et al. (1992) and Silvapulle and Choi (1999) find one-way Granger causality from volume to volatility in US and Korean stock markets respectively but Lee and Rui (2000) and Lee and Rui (2002) do not find trading volume Granger cause returns in Chinese and Japanese markets respectively. Yet, Chan et al. (1996), Jones et al. (1999), Giot (2010) motivate another strand of the empirical literature on volume volatility comovements and question if trading volume measured as such is indeed the driver of the volume volatility relation for individual securities.

securities in multiple markets? How volume modifies volatility persistence in individual securities and markets? And finally, how volume impacts volatility persistence and transmission across cross listed securities?

Gagnon and Karolyi (2006) review the literature on volatility spillover across geographically dispersed markets and report inconclusive evidence with respect to the volatility transmission across country index returns. From a study of multiple developed and emerging markets, Diebold and Yilmaz (2009) observe fundamental differences between return and volatility spillover- return spillover shows trend but no busts while volatility shows no trend and only busts.

The issue of volume effect on volatility persistence has been studied, albeit only cursorily and in the context of market indexes, by Chen et al. (2001), Girard and Biswas (2007), and Vlastakis and Markellos (2012). Those studies report that the coefficient associated with lag country index volatility persistence parameter do not alter after adding volume into a volatility model.

On the other hand, the empirical evidence with respect to volatility spillover for dually listed securities is limited, mostly country specific, and scattered. Chan et al. (1996) compare the intraday volume, volatility, and spread of Japanese and European ADRs with those of the underlying securities and Hauser et al. (1998) investigate the information flow between NYSE and Tel Aviv exchange for five stocks listed in both exchanges. Yang (2007) and Chen et al. (2010) model return and volatility spillover for Japanese and Chinese ADRs respectively. Following up on Karolyi (1995), Xu and Fung (2002) and Poshakwale and Aquino (2008) fit bivariate GARCH models for

Chinese ADRs, and a global basket of ADRs and their underlying stocks respectively where only the volatility functions are interrelated.

### **3. Institutional Structure of NYSE Foreign Listings, Data, and Preliminary Results**

#### *3.1 Data description and Institutional structure of ADR Listings*

As of December 31, 2012, 525 common and preferred equities from approximately 500 foreign issuers in 46 countries are listed on the NYSE Market among which 259 are traded as ADRs (including Global Shares and NY Registry Shares) with primary listing in an exchange located outside the US.<sup>6</sup> Below, we briefly discuss the institutional details of the ADR listing process at the NYSE and the supporting ADR database to motivate the empirical framework adopted in this paper.

A foreign issuer may choose to list on NYSE Market, which includes the main exchange, NYSE for established mid to large cap companies and an electronic exchange without a designated specialist for relatively new and smaller companies. NYSE Market listing allows foreign corporations to market their securities in the form of depositary receipts known as ADRs to US investors; on the contrary, it allows US investors to invest directly and domestically in shares of foreign corporations. Often, those foreign corporations also raise capital in the US via IPO or SEO. While at the

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<sup>6</sup> Among the 525 NYSE and NYSE MKT listed ADRs from 46 countries, a majority of recently listed ADRs from China and also a few other emerging markets listed on NYSE MKT do not have a primary listing. All issuers from Canada and a few small nations/territories along with a majority of issuers from China have issued original listings (ORD) instead of ADRs (ORDs do not require sponsorship by a depositary bank- for more details, please refer to <https://research.scottrade.com/knowledgecenter/Public/help/Article?docId=37010be1721740e0879fb4b3510db8ed>); incidentally, those are also the two countries with the highest number of foreign issuers at the NYSE market. Global and NY registry shares are for securities from Netherlands and Luxembourg, which are prohibited by domestic currency laws to sell shares in another currency.

beginning of the ADR program, only large and reputable foreign companies would list and raise capital in the US, since the mid-2000, many small unlisted foreign companies have issued ADRs and raised capital in the US via IPOs. Prior listing in its home or another foreign jurisdiction is not mandatory for a foreign issuer and hence many foreign issuers, albeit only recently, have bypassed their domestic capital markets and resorted to simultaneously issuing IPOs and ADRs in the USA; many of those foreign issuers have subsequently raised capital and listed on a primary exchange in their respective home countries.

Table 1 provides a breakdown of our sample of 170 ADRs and their corresponding home country shares listed in 29 foreign exchanges located in 17 developed and 12 emerging markets (MSCI Developed and Emerging Markets Indexes include 16 and 11 of those markets respectively) respectively.<sup>7</sup> Appendix A contains our sample ADRs and their underlying home securities with other relevant details.

### *3.2 Descriptive statistics on ADR and their corresponding home security volume and return*

We obtain daily data on closing prices, number of shares traded, and the value of traded shares (turnover) for those 170 ADRs and their underlying home securities listed on the respective domestic Stock Exchanges from Bloomberg data terminals for a period of

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<sup>7</sup> Our primary criterion for identifying developed vs. emerging economies is based on MSCI ACWI indexes of 23 Developed and 23 Emerging Markets as of July 2014 as listed in Table 1. As per MSCI ACWI components list, eight developed markets (Austria, Canada, Hong Kong, New Zealand, Singapore, Sweden) and seven emerging markets (Czech Republic, Egypt, Hungary, Malaysia, Qatar, Thailand, and UAE) do not have any ADR listed on the NYSE Market. Further 23 ADRs/home country shares are either preferred shares and/or do not have the necessary data and hence are excluded. Included in those 23 securities are two each from Colombia and Russia and one each from Peru and Turkey, which denote their entire ADR listings respectively and are therefore completely eliminated from the sample due to missing data. Hence our final sample consists of 142 securities from 26 developed and emerging markets.

approximately eight to 22 years, from the initial trading day for each security beginning on or after April 1983. Hence, the sample period for each pair of ADR and its underlying security begins from their respective initial trading dates from/after April 1, 1983 and ends on December 31, 2013. We also obtain similar data on prices and volume, and exchange rates for those 29 developed and emerging markets, where our sample securities have their primary listings.

We denote volume  $v_t^i = \ln(N_t^i)$  and compute returns as  $r_t^i = \ln\left(\frac{P_t^i}{P_{t-1}^i}\right)$  where  $N(i, t)$  and  $P(i, t)$  denote number of trades and closing price respectively of the  $i$ th security on the  $t$ th day. Table 2 provides descriptive statistics on trading volume and daily returns for our sample ADRs and their underlying securities listed on home exchanges. It also provides country wise summaries of those statistics. The descriptive statistics indicate wide differences between means of daily returns for both ADRs and their corresponding home securities, albeit some of the differences seem to be due to the varying length of each security's time series. In fact, while most ADRs show positive mean daily returns, most home securities show negative mean day returns during the sample period. A few exceptions are ALU and TOT (France), all three securities (AEG, ING, PHG) in The Netherlands, and three seemingly outlier securities (BBVA, SAN, TEF) in Spain. Standard deviations of returns are similar for ADRs and their underlying home shares. Nevertheless, there are visible differences among the mean returns and standard deviations on a country by country basis.

In contrast, for almost all securities, home shares trading volumes are demonstrably much larger than that of ADRs. There are 23 exceptions, for example,

VIV and VALE (Brazil), ELN (Ireland), and TEVA (Israel) where the ADR volumes are higher than those of home shares. On the contrary, BCA's (Chile) average trading volume of home shares is 8572 times its average volume at NYSE. Among countries, only Brazil and Israel show average trading volumes in ADRs higher than that of home shares. Japan (345 times) and China/Hong Kong (198 times) have the two highest ratios of average volume in home shares to average volume in ADR.

The top panel in Figure 1 graphically shows the distribution of risk denoted by standard deviation against mean return for daily return for ADRs and their underlying home shares, which seem to indicate almost similar risk return profile for ADRs and their corresponding home shares.

We conduct formal tests of joint mean variance equality for our sample ADRs and their corresponding home shares. In spite of the seemingly similar distributions of returns for ADRs and their underlying home shares, the test results for the equality of distributions based on means and variances between daily returns reject the null hypothesis for a large number of ADRs and their corresponding home securities. We also find significant differences between the distributional properties of day and daily returns, which seem to indicate bias in the results of those studies which combine day and daily returns to account for large time differences between markets, which do not have overlapping market open hours. He and Yang (2012) report differences between daytime and overnight returns spillover for Chinese ADRs and their home shares.

### 3.3. Trading volume expectation model and volume surprise

We denote volume by the number of shares traded and decompose it into expected and unanticipated (surprise) components as follows.<sup>8</sup> For each pair of ADR and its underlying home share in our sample, we use the base model without any of the control variables from Halling et al. (2013) as in 1a-1b below to empirically estimate trading volume forecast based on a VAR methodology.<sup>9</sup>

$$v_{i,t}^{us} = \alpha + \beta v_{i,t-1}^{us} + \phi v_{t-1}^{us} + V_{i,t}^{us} \quad (1a)$$

$$v_{i,t}^h = \alpha + \mu_i v_{i,t-1}^h + \theta_j v_{t-j}^h + V_{i,t}^h \quad (1b)$$

Table 3 contains country wise summaries of the parameter estimates from models 1a and 1b, and ADF statistics on volume surprises, the error terms in equations 1a and 1b for our sample securities. The VAR parameters are OLS estimates with Newey-West correction for autocorrelation and heteroscedasticity. The t-statistics corresponding to the parameters and the model fit statistics given by F-stats and R<sup>2</sup> provide strong and concrete evidence of volume comovements for ADRs and their underlying home shares. All ADRs and their underlying home shares show consistently significant (at less than 1%) negative 1<sup>st</sup> order auto correlation; however, with regards to cross correlation, while almost all home shares also show significant (at less than 1%) 1<sup>st</sup> order positive cross correlations in volume, ADRs show somewhat mixed evidence in terms of cross volume correlations. Regarding cross security volume

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<sup>8</sup> We also use turnover as a measure of volume.

<sup>9</sup> Adjusting individual security volume with market volumes creates asymmetric effects between active and thinly traded stocks. Liquid stocks may suffer a larger impact of market volume changes than their thinly traded counterparts. This asymmetry may spill over to home and/or host markets in a largely unpredictable way since while ADR volumes are customarily low in the US, a majority of ADR issuers are among the most active stocks in the respective markets.



comovements for ADRs, the numbers of positive (negative) significant (at least lower than 10%) 1<sup>st</sup> order cross correlation in volume are 24 (28) and hence not only the direction of the coefficients are inconsistent but also a majority of the ADRs do not display cross security volume effect. This evidence of one sided, US to home, effect is observed in cross country returns but given in terms of volume, US accounts for a smaller proportion of volume than that accruing from the home market, is surprising nonetheless. The lack of consistency in the cross volume serial correlation coefficients, mainly for ADRs do not seem to be explained by obvious differences among home countries, for example, developed vs. emerging markets or time zone effects. The error terms from each of the equations show unanticipated volume or volume surprise denoted by  $V_t$  and how it affects security returns and volatility spillover is the fundamental issue we investigate in this paper.

#### **4. Testing hypotheses on the effect volume surprise on return spillover**

##### *4.1 Effect of volume surprise on return spillover*

We estimate the return autocorrelation and cross correlation parameters as in Gagnon and Karolyi (2009) from the following set of bivariate VAR models, where  $r_{i,t}^{us}$  and  $r_{i,t}^h$  denote daily returns on the  $i$ th ADR or the home security at time  $t$ . For each security, we include two interaction terms, trading volumes \* returns corresponding to each ADR and its underlying home security, which denote the respective volume effects on lag return comovements. We call these volume induced spillover effects. The resulting

models are as follows.

$$r_{i,t}^{us} = a_1 + b_1 r_{i,t-1}^{us} + c_1 r_{i,t-1}^h + d_1 r_{i,t-1}^{us} V_{i,t-1}^{us} + e_1 r_{i,t-1}^h V_{i,t-1}^h + f_1 r_{t-1}^{us} + g_1 r_{t-1}^h + e_{i,t}^{us} \quad (2a)$$

$$r_{i,t}^h = a_2 + b_2 r_{i,t-1}^h + c_2 r_{i,t-1}^{us} + d_2 r_{i,t-1}^h V_{i,t-1}^h + e_2 r_{i,t-1}^{us} V_{i,t-1}^{us} + f_2 r_{t-1}^h + g_2 r_{t-1}^{us} + e_{i,t}^h \quad (2b)$$

The parameter vectors  $b_i$  and  $c_i$  where  $i=\{1,2\}$  in equations 2a-2b signify time dependence and predictability of returns via serial auto and cross correlations while the coefficients  $d_1$ ,  $d_2$ ,  $e_1$ , and  $e_2$  denote return reversals or continuation based on last period trading volume in ADR and home security respectively. The null hypotheses with respect to parameters of the VAR equations 2a-2b  $b_i = c_i = d_i = e_i = 0$  for all  $i > 0$ , where  $i=\{1,2\}$  indicate full and absolute international equities market integration in which neither any short term memory nor a cross market shock may predict future returns on ADRs and their underlying home securities. Based on Blume et al. (1994), Wang (1993), and Campbell et al. (1993), we expect  $b_i \geq 0$  and  $c_i \geq 0$ , while  $d_i \neq 0$  and  $e_i \neq 0$ .

For parameter estimation, we use four different variations of the VAR model in equations 2a-2b. Model 1 is the base model in equations 2a-2b; Model 2 adds a binomial dummy separating between positive and negative volume residual\* return; Model 3 further refines model 2 and includes a trinomial dummy variable to separate volume residual\*return into three groups- a) positive volume shock with positive return b) negative volume shock with negative return c) positive (negative) volume shock with negative (positive) return; Model 4 uses a 50 days MA filter to determine volume residual. We use model 4 for comparing our results with those of Gagnon and Karolyi (2009), who use a similar 50 days MA filter to extract volume residual.

Table 4 contains two sets of results for models 1-4 of the VAR model in equations

2a-2b. First, the summary statistics for models 1-4 for the entire sample are reported in the bottom four rows; the top section of the table reports country wise breakdown of the parameter estimates for model 3. The results indicate significant negative (positive) lagged dependence for ADR and its underlying home security returns; however, regarding cross security dependence, all home securities but only a handful of ADRs display lag cross dependence. The lagged dependence, irrespective of whether it is auto or cross dependence is inconsistent for both ADRs and underlying securities when volume innovation is included. In fact, for volume induced spillover, the split between positively and negatively significant parameters is just around 1/3<sup>rd</sup> each, leaving the remaining 1/3<sup>rd</sup> to zero (insignificant) effect. In the next section, we use this observation to break up the spillover parameters into three separate groups, + significant, - significant, and insignificant for the purpose of identifying determinants of those volume induced spillover parameters.

#### *4.2 Determinants of the impact of volume surprise on return spillover*

We estimate the parameters of the following SUR model.

$$D_t^1 = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1 \quad (3a)$$

$$E_t^1 = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1 \quad (3b)$$

$$D_t^2 = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1 \quad (3c)$$

$$E_t^2 = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1 \quad (3d)$$

where, the dependent variables are the absolute values of the corresponding parameter ( $d_1$ ,  $d_2$ ,  $e_1$ , and  $e_2$ ) estimates in equations 2a-2b while the independent variables are defined as below and the error terms are correlated.

$X_1 = \text{Ln (MVE)}$

$X_2 = \text{Mktdummy}$  {1 if emerging; 0, if developed}

$X_3 = \text{Condummy}$  {1, if Asia-Pacific; -1, if Europe; 0, if Americas}

$X_4 = \text{Timeolap}$  {1, if overlap; 0, if no overlap}

$X_5 = \text{Lawdummy}$  {1, if English common law; -1, Civil law; 0, if neither}

$X_6 = \text{Parasigndummy}$  {1, if + significant; -1, if – significant; 0, if not significant}

Note that in the above model, other than market capitalization and  $X_6$ , an endogenous security and dependent variable specific trinary choice variable, all other independent variables are exogenous and market/country specific and hence identical for each security within a home country, region, or continent. Hence for this system of equations, the errors are independent and the SUR and OLS estimates are exactly identical. In order to setup a SUR system with at least one unique independent variable for each equation that would allow us to obtain more efficient GLS estimates, we use a two-step process to modify the SUR model. First, we run OLS regressions with each country/market indicator variables to determine significant indicator variable for each dependent variable, the parameter in question. We use  $X_6$  as a control variable in each of those single equation regression models in 3a-3d. Next, we setup a system of equations in which we choose only the significant indicator variable(s) for each equation and one or more interaction terms combining a security specific variable with a significant indicator variable(s).

Table 5 reports the expected values of the dependent variable in question, in this case, the volume induced spillover coefficients noted in equations 3a-3d above, conditional on the binary/trinary choices implicit in the indicator variable. The four columns contain the conditional expected absolute values of the four parameters,  $d_1$ ,  $d_2$ ,

$e_1$ , and  $e_2$  from equations 2a-2b respectively; each row contains the expected absolute value of the column variable corresponding to the condition, the values of the binary/trinary indicator variable. For example, the row corresponding to  $E(Y|X_2=0, X_6=0)$  finds the expected absolute values of  $d_1$ ,  $d_2$ ,  $e_1$ , and  $e_2$  for those developed market securities with insignificant volume induced spillover coefficients to be .022, .019, .050, and .057 (significant at 1% level) respectively. To arrive at the conditional expected values displayed in any cell in subsequent rows below the top row in each block, the respective cell content is added to the target cell in the top row heading a block. For example,  $E(D_1|X_2=1, X_6=0) = .022+.012= .034$ .

The cell contents in Table 5 underscore the differences between the conditional expected absolute values of and hence the varied impact of the indicator variables on the spillover parameters for ADRs ( $d_1$ ,  $e_1$ ) and those of their underlying home securities ( $d_2$ ,  $e_2$ ). Specifically, while the ‘sign’ trinary indicator variable denoting whether any spillover parameter is + significant, - significant, or insignificant is strongly significant (at 5% or lower) for both ADR parameters  $d_1$ ,  $e_1$  denoting higher (lower) absolute values for those parameters for positive (negative) values of the coefficients compared to the coefficients for those securities which are insignificant. However, for home shares only the absolute values of the cross correlation parameter,  $e_2$  is significantly higher (lower) than that of the insignificant coefficient group but the serial autocorrelation parameter,  $d_2$  is either not significant or only weakly significant (at 10% or lower). Hence, while the magnitude of continuation or reversals of serial autocorrelation in returns is higher and more sustainable for ADRs than home shares,

similar magnitude of continuation or reversals of serial cross correlation is higher for home shares than for ADRs. This finding is consistent across any other split in the sample, namely, developed vs. emerging markets, civil vs common law, and others.

Similarly, with regard to the indicator variables,  $X_3$  (continent),  $X_4$  (time overlap), and  $X_5$  (legal origin), there is much asymmetry between ADRs and their underlying home shares. The only exception is  $X_2$  (emerging vs. developed markets) - the mean absolute values of all the four spillover parameters are not statistically different between those of emerging and developed markets securities. Our findings confirm Gagnon and Karolyi (2009) who consider GDP growth, accounting/disclosure system, currency control, legal origin, and emerging vs. developed markets as country specific variables and find scattered and inconsistent evidence of their solo impact on volume induced return spillover.

Overall we find strong evidence that each spillover parameter has a unique set of binary/trinary choice variables, which affect its conditional mean. We use the result from Table 5 to setup a customized system of equations with targeted equation specific interaction variables for the purpose of running a SUR model.

## **5. Testing hypotheses on the effect of volume surprise on volatility spillover**

Volatility is not observable and hence a model is necessary for estimating volatility. We use squared errors from equations 2a and 2b to estimate volatility and test volatility spillover and volume effect on volatility spillover in Section 5.1-5.2 below.

### 5.1 Volatility spillover across markets, persistence and volume effect

We begin with parameter estimation and a Granger causality test of cross security volatility spillover based on the following set of equations for volatility.

$$\sigma_t^2(A) = a_0 + \sum a_{1m} \sigma_{t-m}^2(A) + \sum b_{1m} \sigma_{t-m}^2(H) + \varepsilon_{1A} \quad (4a)$$

$$\sigma_t^2(H) = a_0 + \sum a_{2m} \sigma_{t-m}^2(H) + \sum b_{2m} \sigma_{t-m}^2(A) + \varepsilon_{1H} \quad (4b)$$

The parameter vectors  $a_m$  and  $b_m$  in equations 4a-4b signify volatility autocorrelation and cross security volatility spillover. We test  $a_m = b_m = 0$  for all ADRs and their underlying securities in US and home markets respectively. In the VAR framework, rejecting hypotheses  $b_m = 0$  implies causality, i.e. lag cross security volatility causes volatility.

Table 6A contains country wise summary of the parameter estimates of the VAR model in equations 4a-4b and the corresponding Granger causality test of cross security volatility spillover. The test results indicate significant lagged dependence for all ADRs and their underlying home securities volatility. Further, the Granger causality tests (the number of significant Chi-square statistics for each country is reported) indicate almost universal, significant lagged serial cross correlation between ADR and their underlying home securities.

Further, in order to determine the effect of volume on reducing volatility persistence, we modify the above system of equations to the following VAR setup to test for the effect of volume surprise on volatility denoted by  $V(t)$  and  $\sigma^2(t)$  respectively for each ADR and its underlying home security.

$$\sigma_t^2(A) = a_0 + a_{11} \sigma_{t-1}^2(A) + a_{12} V_{t-1}^A + a_{13} \sigma_{t-1}^2(H) + a_{14} V_{t-1}^H + \varepsilon_1 \quad (5a)$$

$$\sigma_t^2(H) = a_0 + a_{21}\sigma_{t-1}^2(A) + a_{22}V_{t-1}^A + a_{23}\sigma_{t-1}^2(H) + a_{24}V_{t-1}^H + \varepsilon_1 \quad (5b)$$

Table 6B contain country wise breakdown of the results from the above hypotheses tests and indicate continued significant lag dependence in auto and cross volatility for both ADR and its underlying home security volatility. However, almost universally there is no dependence on auto or cross volume. Hence, volume surprise does not reduce volatility persistence for either ADRs or their underlying home shares. This finding raises the possibility that volume surprise may not contain ‘information’ related to price discovery but may just reveal transient liquidity shocks to the market.

## 5.2 Volume effect on volatility spillover

Finally, emulating the modeling approach that we used earlier to capture the unanticipated volume effect on return spillover as in equations 2a-2b, we estimate the parameters of equations 6a and 6b below, where  $V_{t-1}$  is one lag volume surprise and return variance  $\sigma_t^2$  estimated as squared errors from equations 2a-2b.

$$\sigma_t^2(A) = a_0 + b_1\sigma_{t-1}^2(A) + d_1V_{t-1}^A\sigma_{t-1}^2(A) + c_1\sigma_{t-1}^2(H) + e_1V_{t-1}^H\sigma_{t-1}^2(H) + \varepsilon_1 \quad (6a)$$

$$\sigma_t^2(H) = a_0 + b_2\sigma_{t-1}^2(A) + d_2V_{t-1}^A\sigma_{t-1}^2(A) + c_2\sigma_{t-1}^2(H) + e_2V_{t-1}^H\sigma_{t-1}^2(H) + \varepsilon_2 \quad (6b)$$

Note that the parameter vectors  $b_i$  and  $c_i$  in equations 6a and 6b signify volatility autocorrelation and cross security volatility spillover, while the parameter vectors  $d_i$  and  $e_i$  in equations 6a and 6b signify volume induced volatility spillover. We test  $b_i = c_i = d_i = e_i = 0$  against the alternate hypotheses,  $d_i \geq 0$  and  $e_i \geq 0$  for all  $i > 0$ , where  $i = \{1, 2\}$  denote US and home markets respectively, which are based on the empirical implications in Wang (1993, 1994) and Campbell et al. (1993) and tested by Abramov



et al. (2006) in the context of US domestic securities.

Table 7 contains country wise breakdown of the mean spillover parameter estimates; further, the last row contains a sample summary along with the number of positive and negative significant parameters. Evidently, the sample means of all the four volume induced spillover parameters,  $d_1$ ,  $d_2$ ,  $e_1$ , and  $e_2$  are all negative following similar negative coefficients for a majority of the individual securities although  $d_2$  seems to have a relatively higher number of positively (49 positive and 92 negative) significant coefficients.

### *5.3 Determinants of the impact of volume surprise on volatility spillover*

We re-estimate OLS parameters of the system of equations in 3a-3d henceforth replacing the dependent variables to the volume induced spillover parameter estimates from equations 3a-3b by those from equations 6a-6b, instead. Note that we consider each of the binary/trinary, market/country variables  $X_2 - X_5$  along with the control variable,  $X_6$  and estimate the parameters of four single equations for each of the four dependent variables, the volume induced volatility spillover parameters.

Table 8 (similar to Table 5 for return spillover) contains conditional expected absolute values of the spillover parameters,  $d_1$ ,  $d_2$ ,  $e_1$ , and  $e_2$  given the target choice with respect to country/market specific indicator variables,  $X_2 - X_5$  and  $X_6$  as a control variable. The results reported in Table 8 indicates the following. For ADR spillover parameters,  $d_1$ ,  $e_1$  the mean absolute values are higher (lower) for positive (negative) significant parameters than those of insignificant parameters; for home securities, there

is no statistical difference between the means. Further, mean absolute values of  $d_1$ , the autocorrelation parameter for ADR are not statistically different between emerging and developed markets, among different continents, between with or without time overlap, and between common and civil law countries; mean absolute values of  $e_1$ , cross security correlation parameters are not statistically different between emerging and developed markets, among different continents, but higher, albeit weakly significant at 10%, for securities in countries with time overlap and in common law countries than those in countries with no time overlap and in civil law countries respectively. In contrast, the home security spillover parameters,  $d_2$ ,  $e_2$  are higher for developed markets, higher (lower) for Asia-Pacific (Europe) but not statistically different between with or without time overlap and common and civil law countries.

## 6. Conclusion

We empirically investigate the effect of volume on return and volatility spillover for 142 ADRs and their underlying securities listed in 26 developed and emerging market countries. We find that a) cross security lag volume determines security volume in multi market trading; b) a lag interaction term residual volume\*return casts asymmetric impact on return for ADRs and their underlying home securities; c) a lag interaction term residual volume\*volatility casts asymmetric impact on volatility for ADRs and their underlying home securities. Our findings indicate short term return and volatility predictability in cross listed multi market trading shares and that volume induced return and volatility spillover are asymmetric on multiple dimensions. We find that there are

asymmetries between ADRs and their underlying home shares, between return and volatility spillovers, and also between auto and cross correlation parameters. Further those ‘asymmetries’ depend asymmetrically on any number of country/market level indicators but more consistently on security level indicators, for example size and liquidity.

Pending work-in-progress related to this study includes

- a) Updating the database to include missing securities and also non-MSCI countries, namely Argentina and Luxembourg;
- b) Refining the system of equations for SUR estimation by customizing each equation including only significant country/market security indicator variables and security specific proxies for liquidity;
- c) Estimating the parameters of a system of panel data equations as in 3a-3d accounting for time varying estimates of the determinants of volume induced return and volatility spillover.

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Table 1: Sample breakdown into countries included in MSCI ACWI index, which is composed of 23 developed and 23 emerging markets indexes as below.

|                                                                          | Total    | MSCI     | Sample  |
|--------------------------------------------------------------------------|----------|----------|---------|
| Nos. of non-US issuers (countries) listed on NYSE Market as of Dec 2012  | 525 (46) |          |         |
| Nos. of ADRs (countries with $\geq 1$ ADR) listed on NYSE as of Dec 2012 | 259 (33) |          |         |
| ADRs (Countries) included in Developed Markets                           |          | 81 (16)  | 84 (17) |
| Australia                                                                |          | 6        | 6       |
| Belgium                                                                  |          | 2        | 2       |
| Denmark                                                                  |          | 1        | 1       |
| Finland                                                                  |          | 1        | 1       |
| France                                                                   |          | 7        | 7       |
| Germany (1 preferred)                                                    |          | 4        | 3       |
| Ireland                                                                  |          | 3        | 3       |
| Israel                                                                   |          | 2        | 2       |
| Italy                                                                    |          | 4        | 4       |
| Japan                                                                    |          | 15       | 15      |
| Netherlands (3 NY Registry Shares)                                       |          | 5        | 2       |
| Norway                                                                   |          | 1        | 1       |
| Portugal                                                                 |          | 1        | 1       |
| Spain                                                                    |          | 3        | 3       |
| Switzerland (1 NY Registry Shares)                                       |          | 5        | 4       |
| UK                                                                       |          | 23       | 23      |
| Luxembourg*                                                              |          |          | 3       |
| ADRs (Countries) included in Emerging markets                            |          | 148 (15) | 86 (12) |
| Brazil (1 preferred)                                                     |          | 25       | 16      |
| Chile (1 preferred)                                                      |          | 12       | 10      |
| China                                                                    |          | 60       | 12      |
| Colombia                                                                 |          | 2        |         |
| Greece                                                                   |          | 1        | 1       |
| India                                                                    |          | 8        | 6       |
| Indonesia                                                                |          | 1        | 1       |
| Mexico (1 Global Share)                                                  |          | 14       | 12      |
| Peru                                                                     |          | 2        |         |
| Philippines                                                              |          | 1        | 1       |
| Russia                                                                   |          | 2        |         |
| South Africa                                                             |          | 7        | 6       |
| South Korea                                                              |          | 8        | 7       |
| Taiwan                                                                   |          | 5        | 5       |
| Turkey                                                                   |          | 1        |         |
| Argentina*                                                               |          |          | 10      |

Table 2: The left panel contains descriptive statistics on daily trading volume,  $v_t^i = \ln(N_t^i)$  on the  $t$ th day for each of our sample ADR listed on NYSE and its underlying home share listed on the respective home country stock exchange. The right panel includes descriptive statistics on daily return computed as  $r_t^i = \ln(P_t^i / P_{t-1}^i)$  where  $P_t^i$  denotes closing price of security  $i$  on the  $t$ th day, on those ADRs and their underlying shares listed on their respective home country stock exchanges. The row containing country averages are in block letters.

|                  | ADR              | Volume           | Home             | Volume           | ADR          | Return       | Home          | Return       |
|------------------|------------------|------------------|------------------|------------------|--------------|--------------|---------------|--------------|
|                  | Mean             | St. Dev          | Mean             | St. Dev          | Mean         | St. Dev      | Mean          | St. Dev      |
| AWCMY            | 106494.6         | 217794.7         | 8629968.3        | 7648495.4        | 0.000        | 0.030        | -0.015        | 0.030        |
| BHP              | 1651681.8        | 2108531.9        | 12860506.1       | 8298379.8        | 0.000        | 0.024        | -0.008        | 0.021        |
| JHX              | 3318.2           | 9742.3           | 2084845.4        | 1499590.8        | 0.001        | 0.028        | -0.014        | 0.027        |
| OBTLY            | 3170.6           | 9041.3           | 112548.7         | 374384.9         | -0.003       | 0.055        | -0.016        | 0.064        |
| SSN              | 729961.8         | 1194819.6        | 7683065.8        | 22561605.4       | -0.001       | 0.085        | -0.029        | 0.081        |
| WBC              | 87878.7          | 112997.7         | 5885210.6        | 4080325.1        | 0.000        | 0.021        | -0.008        | 0.017        |
| <b>Australia</b> | <b>430417.6</b>  | <b>608821.3</b>  | <b>6209357.5</b> | <b>7410463.6</b> | <b>0.000</b> | <b>0.041</b> | <b>-0.015</b> | <b>0.040</b> |
| BUD              | 1147420.0        | 932431.6         | 2179175.4        | 1272326.2        | 0.001        | 0.016        | -0.009        | 0.016        |
| DEG              | 131390.9         | 118219.1         | 388267.7         | 264123.5         | 0.000        | 0.022        | -0.012        | 0.025        |
| <b>Belgium</b>   | <b>639405.5</b>  | <b>525325.4</b>  | <b>1283721.6</b> | <b>768224.8</b>  | <b>0.000</b> | <b>0.019</b> | <b>-0.011</b> | <b>0.020</b> |
| BBD              | 7831290.7        | 5413923.4        | 611502.8         | 2170851.2        | 0.001        | 0.030        | -0.011        | 0.025        |
| BSBR             | 6463344.1        | 6028380.0        | 2876502.8        | 3306821.3        | -0.001       | 0.023        | -0.016        | 0.023        |
| CIG              | 2296613.0        | 1802382.3        | 145640.6         | 176815.6         | 0.001        | 0.030        | -0.014        | 0.028        |
| CPL              | 394862.3         | 298425.8         | 841843.7         | 1001381.7        | 0.000        | 0.024        | -0.014        | 0.022        |
| EBR              | 868216.8         | 649703.2         | 1472277.3        | 1187710.0        | -0.001       | 0.032        | -0.018        | 0.032        |
| ERJ              | 769524.0         | 589715.1         | 1082098.9        | 1193192.7        | 0.000        | 0.028        | -0.017        | 0.031        |
| FBR              | 799096.0         | 854860.9         | 711720.5         | 815748.1         | 0.000        | 0.031        | -0.018        | 0.033        |
| GFA              | 2222667.2        | 1460041.5        | 6755572.5        | 6374278.7        | -0.001       | 0.046        | -0.027        | 0.048        |
| OIBR             | 808798.2         | 1468041.4        | 204567.3         | 751249.3         | 0.000        | 0.037        | -0.011        | 0.036        |
| PBR              | 11578084.3       | 8741422.2        | 4177294.0        | 3311323.6        | 0.000        | 0.030        | -0.015        | 0.029        |
| SBS              | 1557926.4        | 1197217.6        | 1197656.2        | 873092.0         | 0.001        | 0.030        | -0.017        | 0.029        |
| SID              | 4449706.7        | 3792935.1        | 4402944.9        | 3101419.8        | 0.000        | 0.036        | -0.018        | 0.036        |
| TSU              | 602587.8         | 672115.9         | 1180677.4        | 2057492.8        | 0.000        | 0.035        | -0.023        | 0.042        |
| UGP              | 274444.1         | 290466.5         | 610171.5         | 645401.6         | 0.001        | 0.026        | -0.011        | 0.024        |
| VALE             | 17798648.6       | 12275039.0       | 2863466.7        | 1933156.4        | 0.001        | 0.030        | -0.014        | 0.028        |
| VIV              | 435350.5         | 779685.3         | 42049.6          | 138546.2         | 0.000        | 0.027        | -0.011        | 0.026        |
| <b>Brazil</b>    | <b>3696947.5</b> | <b>2894647.2</b> | <b>1823499.2</b> | <b>1814905.1</b> | <b>0.000</b> | <b>0.031</b> | <b>-0.016</b> | <b>0.031</b> |
| BCA              | 30538.0          | 76470.7          | 261778399.9      | 345751579.6      | 0.000        | 0.022        | -0.010        | 0.019        |
| BCH              | 16492.1          | 22097.6          | 29323638.9       | 62441199.0       | 0.001        | 0.021        | -0.009        | 0.019        |
| BSAC             | 312371.0         | 467508.7         | 64027850.4       | 117768540.9      | 0.000        | 0.020        | -0.012        | 0.023        |
| CCU              | 115006.4         | 209918.6         | 352367.3         | 1472685.6        | 0.000        | 0.019        | -0.010        | 0.021        |
| ENI              | 368087.4         | 450920.1         | 22815900.3       | 31187131.4       | 0.000        | 0.022        | -0.014        | 0.026        |
| EOC              | 142524.2         | 225169.9         | 6411800.0        | 6600765.9        | 0.000        | 0.020        | -0.012        | 0.022        |

|                  | ADR               | Volume            | Home               | Volume             | ADR          | Return       | Home          | Return       |
|------------------|-------------------|-------------------|--------------------|--------------------|--------------|--------------|---------------|--------------|
|                  | Mean              | St. Dev           | Mean               | St. Dev            | Mean         | St. Dev      | Mean          | St. Dev      |
| LFL              | 364776.8          | 446224.1          | 531140.6           | 1124445.2          | 0.000        | 0.026        | -0.013        | 0.030        |
| PVD              | 22688.4           | 90378.3           | 348933.6           | 914631.9           | 0.000        | 0.018        | -0.006        | 0.019        |
| SQM              | 458118.1          | 865570.0          | 360156.7           | 435979.3           | 0.000        | 0.024        | -0.014        | 0.032        |
| VCO              | 25391.5           | 196190.0          | 469215.3           | 927888.3           | 0.000        | 0.019        | -0.007        | 0.021        |
| <b>Chile</b>     | <b>185599.4</b>   | <b>305044.8</b>   | <b>38641940.3</b>  | <b>56862484.7</b>  | <b>0.000</b> | <b>0.021</b> | <b>-0.011</b> | <b>0.023</b> |
| NVO              | 707226.5          | 1045171.1         | 6854887.1          | 5973360.3          | 0.001        | 0.020        | -0.011        | 0.023        |
| <b>Denmark</b>   | <b>707226.5</b>   | <b>1045171.1</b>  | <b>6854887.1</b>   | <b>5973360.3</b>   | <b>0.001</b> | <b>0.020</b> | <b>-0.011</b> | <b>0.023</b> |
| NOK              | 18793417.6        | 20495147.3        | 23792152.7         | 20913047.6         | 0.000        | 0.033        | 0.178         | 0.561        |
| <b>Finland</b>   | <b>18793417.6</b> | <b>20495147.3</b> | <b>23792152.7</b>  | <b>20913047.6</b>  | <b>0.000</b> | <b>0.033</b> | <b>0.178</b>  | <b>0.561</b> |
| ALU              | 6465283.9         | 10537602.9        | 16225268.0         | 16759945.8         | 0.000        | 0.037        | 0.248         | 0.663        |
| ORAN             | 420884.6          | 593845.1          | 8715665.8          | 6184855.9          | 0.000        | 0.027        | 0.120         | 0.487        |
| SNY              | 1668879.6         | 1646706.2         | 3794283.0          | 2308165.1          | 0.000        | 0.019        | -0.011        | 0.020        |
| TOT              | 1526951.7         | 1511071.7         | 7193160.5          | 5065749.2          | 0.000        | 0.019        | 0.259         | 0.660        |
| VE               | 219704.1          | 318086.7          | 2170266.1          | 1512829.4          | 0.000        | 0.025        | -0.015        | 0.028        |
| <b>France</b>    | <b>2060340.8</b>  | <b>2921462.5</b>  | <b>7619728.7</b>   | <b>6366309.1</b>   | <b>0.000</b> | <b>0.025</b> | <b>0.120</b>  | <b>0.372</b> |
| FMS              | 167043.4          | 373007.7          | 905551.0           | 723468.2           | 0.000        | 0.021        | -0.013        | 0.024        |
| <b>Germany</b>   | <b>167043.4</b>   | <b>373007.7</b>   | <b>905551.0</b>    | <b>723468.2</b>    | <b>0.000</b> | <b>0.021</b> | <b>-0.013</b> | <b>0.024</b> |
| ACH              | 597633.3          | 720383.4          | 37932556.0         | 37154771.4         | 0.000        | 0.036        | -0.019        | 0.039        |
| CEA              | 41158.5           | 77681.8           | 19465710.4         | 24965697.9         | 0.000        | 0.039        | -0.025        | 0.046        |
| CHA              | 174718.7          | 182477.6          | 106509696.1        | 86342699.5         | 0.000        | 0.028        | -0.016        | 0.028        |
| GSH              | 38784.8           | 50180.7           | 7666724.2          | 8566020.7          | 0.000        | 0.029        | -0.017        | 0.033        |
| HNP              | 201151.4          | 190009.5          | 26051297.5         | 16947826.3         | 0.000        | 0.027        | -0.016        | 0.029        |
| LFC              | 1000097.5         | 1797104.9         | 70616901.3         | 63346083.6         | 0.001        | 0.028        | -0.014        | 0.028        |
| PTR              | 433895.5          | 456955.7          | 122044955.5        | 117598760.7        | 0.001        | 0.025        | -0.013        | 0.027        |
| SHI              | 36530.3           | 46800.0           | 21895148.9         | 23261070.6         | 0.000        | 0.036        | -0.023        | 0.042        |
| SMI              | 252630.4          | 353860.4          | 89293622.2         | 122292247.7        | -0.001       | 0.038        | -0.022        | 0.042        |
| SNP              | 357121.0          | 368477.1          | 144423359.2        | 132229071.2        | 0.001        | 0.027        | -0.015        | 0.029        |
| YZC              | 261378.4          | 308914.1          | 20781747.1         | 15142717.3         | 0.000        | 0.038        | -0.021        | 0.043        |
| ZNH              | 58610.4           | 90933.6           | 17858700.0         | 16160609.0         | 0.000        | 0.039        | -0.024        | 0.045        |
| <b>Hong Kong</b> | <b>287809.2</b>   | <b>386981.6</b>   | <b>57045034.9</b>  | <b>55333964.7</b>  | <b>0.000</b> | <b>0.032</b> | <b>-0.019</b> | <b>0.036</b> |
| HDB              | 1198820.4         | 1229344.1         | 3453627.5          | 4552431.9          | 0.001        | 0.029        | -0.015        | 0.025        |
| IBN              | 1462057.1         | 1576927.0         | 3453846.8          | 5268845.1          | 0.000        | 0.037        | -0.021        | 0.037        |
| INFY             | 1930013.3         | 1720751.0         | 1388641.7          | 1608781.2          | 0.000        | 0.026        | -0.009        | 0.027        |
| RDY              | 355174.5          | 333328.2          | 433731.1           | 395660.8           | 0.001        | 0.024        | -0.017        | 0.026        |
| TTM              | 1281029.4         | 996650.2          | 15118853.1         | 12713395.4         | 0.001        | 0.032        | -0.021        | 0.037        |
| WIT              | 831134.6          | 705128.0          | 2727297.1          | 2376250.5          | 0.000        | 0.033        | -0.021        | 0.036        |
| <b>India</b>     | <b>1176371.6</b>  | <b>1093688.1</b>  | <b>4429332.9</b>   | <b>4485894.2</b>   | <b>0.000</b> | <b>0.030</b> | <b>-0.017</b> | <b>0.031</b> |
| TLK              | 333484.7          | 427109.8          | 125460933.8        | 119351720.7        | 0.000        | 0.033        | -0.015        | 0.033        |
| <b>Indonesia</b> | <b>333484.7</b>   | <b>427109.8</b>   | <b>125460933.8</b> | <b>119351720.7</b> | <b>0.000</b> | <b>0.033</b> | <b>-0.015</b> | <b>0.033</b> |
| CRH              | 174665.2          | 264259.6          | 1691198.1          | 1247485.4          | 0.000        | 0.028        | -0.017        | 0.031        |
| ELN              | 3937993.8         | 7138215.6         | 317843.1           | 491379.0           | 0.000        | 0.055        | -0.025        | 0.066        |
| IRE              | 149659.6          | 433499.8          | 14882791.7         | 39197450.0         | -0.001       | 0.045        | -0.027        | 0.058        |

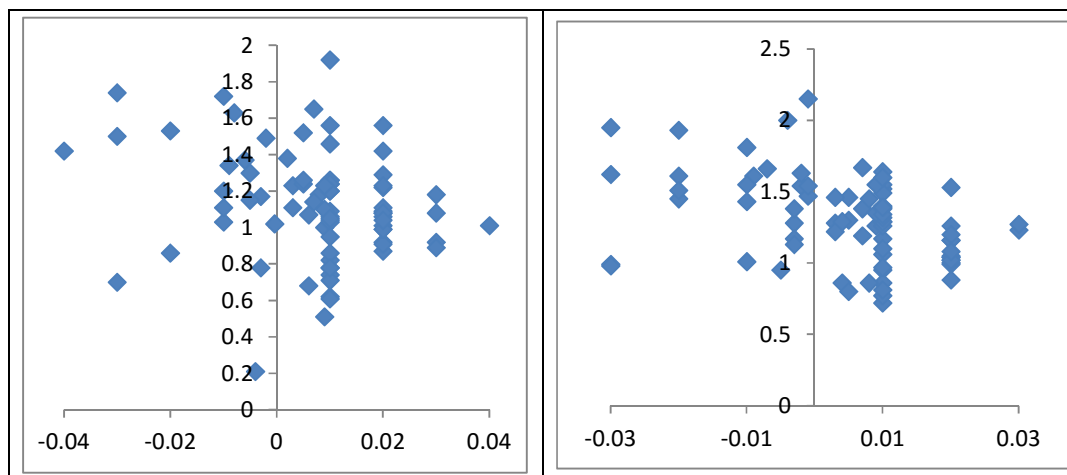
|                    | ADR              | Volume           | Home              | Volume            | ADR          | Return       | Home          | Return       |
|--------------------|------------------|------------------|-------------------|-------------------|--------------|--------------|---------------|--------------|
|                    | Mean             | St. Dev          | Mean              | St. Dev           | Mean         | St. Dev      | Mean          | St. Dev      |
| <b>Ireland</b>     | <b>1420772.9</b> | <b>2611991.6</b> | <b>5630611.0</b>  | <b>13645438.1</b> | <b>0.000</b> | <b>0.043</b> | <b>-0.023</b> | <b>0.052</b> |
| BSI                | 9422.1           | 17346.8          | 49826.6           | 74868.3           | 0.000        | 0.031        | -0.015        | 0.031        |
| TEVA               | 4506775.8        | 3853839.6        | 498157.2          | 339807.3          | 0.000        | 0.014        | -0.006        | 0.017        |
| <b>Israel</b>      | <b>2258098.9</b> | <b>1935593.2</b> | <b>273991.9</b>   | <b>207337.8</b>   | <b>0.000</b> | <b>0.023</b> | <b>-0.011</b> | <b>0.024</b> |
| E                  | 416488.1         | 449037.4         | 19537939.8        | 15758133.7        | 0.000        | 0.019        | -0.010        | 0.020        |
| LUX                | 140186.7         | 155489.3         | 840328.7          | 792943.7          | 0.000        | 0.021        | -0.013        | 0.023        |
| TI                 | 234536.8         | 259966.2         | 108526218.2       | 75386724.6        | 0.000        | 0.024        | -0.014        | 0.024        |
| <b>Italy</b>       | <b>263737.2</b>  | <b>288164.3</b>  | <b>42968162.2</b> | <b>30645934.0</b> | <b>0.000</b> | <b>0.021</b> | <b>-0.012</b> | <b>0.022</b> |
| ATE                | 17808.0          | 41038.9          | 2552081.8         | 1392821.3         | 0.000        | 0.030        | -0.017        | 0.033        |
| CAJ                | 361018.7         | 300616.3         | 5239679.1         | 2631220.2         | 0.000        | 0.021        | -0.011        | 0.024        |
| DCM                | 180133.2         | 219645.4         | 8808326.2         | 5233942.8         | 0.000        | 0.025        | -0.011        | 0.025        |
| HMC                | 378307.9         | 472726.0         | 5400265.8         | 3581320.3         | 0.000        | 0.021        | -0.012        | 0.026        |
| IX                 | 22177.1          | 39522.5          | 8775088.8         | 11349957.3        | 0.000        | 0.030        | -0.016        | 0.035        |
| KNM                | 5794.8           | 14756.3          | 1067598.8         | 769184.9          | 0.000        | 0.026        | -0.015        | 0.030        |
| KUBTY              | 25467.9          | 46852.7          | 4490492.8         | 3447000.3         | 0.000        | 0.026        | -0.015        | 0.031        |
| KYO                | 43821.6          | 57841.3          | 1658150.2         | 1112005.6         | 0.000        | 0.024        | -0.013        | 0.027        |
| MFG                | 504123.9         | 623562.6         | 152123914.6       | 109808812.8       | -0.001       | 0.029        | -0.017        | 0.032        |
| MTU                | 1679461.2        | 1773020.6        | 47726354.2        | 37235254.1        | 0.000        | 0.026        | -0.015        | 0.029        |
| NJ                 | 103520.8         | 194086.5         | 1396761.0         | 882438.4          | 0.001        | 0.025        | -0.013        | 0.027        |
| NMR                | 478477.0         | 821076.7         | 23760659.9        | 26755745.4        | 0.000        | 0.028        | -0.015        | 0.030        |
| NTT                | 267816.0         | 326904.5         | 2104008.1         | 1329373.7         | 0.000        | 0.021        | -0.012        | 0.022        |
| PCRFY              | 233277.7         | 294537.9         | 7061701.1         | 8612922.3         | 0.000        | 0.021        | -0.012        | 0.024        |
| SMFG               | 912327.5         | 1619312.7        | 8919347.1         | 4517789.3         | 0.001        | 0.019        | -0.010        | 0.021        |
| SNE                | 863404.0         | 1226713.9        | 5881011.2         | 6828041.7         | 0.000        | 0.024        | -0.013        | 0.027        |
| TM                 | 430302.8         | 603530.4         | 7906362.9         | 4610259.1         | 0.000        | 0.019        | -0.011        | 0.023        |
| <b>Japan</b>       | <b>382778.8</b>  | <b>510337.9</b>  | <b>17345400.2</b> | <b>13535181.7</b> | <b>0.000</b> | <b>0.024</b> | <b>-0.013</b> | <b>0.027</b> |
| AMX                | 8979495.3        | 5743675.6        | 83661778.6        | 46718095.8        | 0.001        | 0.023        | -0.011        | 0.024        |
| ASR                | 107931.9         | 163372.1         | 367404.5          | 921356.0          | 0.001        | 0.027        | -0.009        | 0.028        |
| CX                 | 6837332.3        | 7699999.0        | 23462304.4        | 25307200.4        | 0.000        | 0.030        | -0.015        | 0.033        |
| FMX                | 848418.6         | 692366.4         | 3374790.3         | 2797439.7         | 0.001        | 0.024        | -0.012        | 0.026        |
| GMK                | 24930.6          | 36393.3          | 733272.2          | 1393031.1         | 0.000        | 0.032        | -0.013        | 0.033        |
| HXM                | 448954.7         | 529627.9         | 1192268.8         | 2146708.5         | -0.001       | 0.038        | -0.019        | 0.046        |
| IBA                | 15062.7          | 21256.9          | 104002.5          | 284652.4          | 0.000        | 0.031        | -0.006        | 0.028        |
| ICA                | 139225.4         | 261751.3         | 1759555.0         | 2696315.8         | 0.000        | 0.036        | -0.018        | 0.037        |
| MXT                | 36546.9          | 223948.6         | 869296.0          | 1178527.8         | -0.001       | 0.043        | -0.026        | 0.043        |
| PAC                | 208488.2         | 522053.4         | 916828.3          | 2051511.9         | 0.000        | 0.023        | -0.012        | 0.023        |
| SIM                | 74588.3          | 154176.4         | 219130.6          | 583841.4          | 0.000        | 0.053        | -0.015        | 0.051        |
| <b>Mexico</b>      | <b>1610997.7</b> | <b>1458965.5</b> | <b>10605511.9</b> | <b>7825334.6</b>  | <b>0.000</b> | <b>0.033</b> | <b>-0.014</b> | <b>0.034</b> |
| AEG                | 546042.5         | 675418.8         | 8044119.2         | 6048141.0         | 0.000        | 0.031        | 0.098         | 0.284        |
| ING                | 1414358.5        | 1793438.9        | 17220268.3        | 11514542.4        | 0.000        | 0.032        | 0.057         | 0.233        |
| PHG                | 1345248.2        | 1301142.2        | 7930283.6         | 5486333.7         | 0.000        | 0.027        | 0.100         | 0.281        |
| <b>Netherlands</b> | <b>1101883.1</b> | <b>1256666.6</b> | <b>11064890.4</b> | <b>7683005.7</b>  | <b>0.000</b> | <b>0.030</b> | <b>0.085</b>  | <b>0.266</b> |

|                    | ADR              | Volume           | Home              | Volume            | ADR          | Return       | Home          | Return       |
|--------------------|------------------|------------------|-------------------|-------------------|--------------|--------------|---------------|--------------|
|                    | Mean             | St. Dev          | Mean              | St. Dev           | Mean         | St. Dev      | Mean          | St. Dev      |
| STO                | 999791.2         | 1046917.7        | 8354798.3         | 7812458.0         | 0.000        | 0.023        | -0.011        | 0.022        |
| <b>Norway</b>      | <b>999791.2</b>  | <b>1046917.7</b> | <b>8354798.3</b>  | <b>7812458.0</b>  | <b>0.000</b> | <b>0.023</b> | <b>-0.011</b> | <b>0.022</b> |
| PHI                | 187740.8         | 170256.7         | 165224.1          | 247417.4          | 0.000        | 0.022        | -0.009        | 0.025        |
| <b>Philippines</b> | <b>187740.8</b>  | <b>170256.7</b>  | <b>165224.1</b>   | <b>247417.4</b>   | <b>0.000</b> | <b>0.022</b> | <b>-0.009</b> | <b>0.025</b> |
| PT                 | 279413.0         | 855945.3         | 4114881.4         | 3795258.7         | 0.000        | 0.023        | -0.013        | 0.024        |
| <b>Portugal</b>    | <b>279413.0</b>  | <b>855945.3</b>  | <b>4114881.4</b>  | <b>3795258.7</b>  | <b>0.000</b> | <b>0.023</b> | <b>-0.013</b> | <b>0.024</b> |
| AU                 | 1387149.4        | 1201112.4        | 805037.0          | 1220785.8         | 0.000        | 0.030        | -0.016        | 0.030        |
| DRD                | 76075.7          | 71771.2          | 944155.8          | 939174.1          | 0.000        | 0.031        | -0.018        | 0.031        |
| GFI                | 2452469.3        | 2869534.9        | 1574433.6         | 2181303.6         | 0.000        | 0.035        | -0.018        | 0.035        |
| HMY                | 1595742.3        | 1552556.5        | 1065287.9         | 1119645.6         | 0.000        | 0.038        | -0.018        | 0.039        |
| SPP                | 148816.5         | 213324.7         | 5373288.5         | 4444745.9         | 0.000        | 0.030        | -0.010        | 0.068        |
| SSL                | 224897.6         | 304103.6         | 1563340.4         | 1274130.8         | 0.000        | 0.028        | -0.014        | 0.029        |
| <b>S. Africa</b>   | <b>980858.5</b>  | <b>1035400.5</b> | <b>1887590.6</b>  | <b>1863297.6</b>  | <b>0.000</b> | <b>0.032</b> | <b>-0.016</b> | <b>0.039</b> |
| KB                 | 310650.0         | 274664.1         | 2040158.6         | 1879930.4         | 0.000        | 0.036        | -0.015        | 0.031        |
| KEP                | 577406.4         | 466587.3         | 1482562.7         | 1197164.5         | 0.000        | 0.028        | -0.015        | 0.028        |
| LPL                | 1091107.2        | 804203.9         | 2439861.9         | 2471130.6         | 0.000        | 0.033        | -0.018        | 0.031        |
| PKX                | 458191.8         | 380110.3         | 290315.9          | 301869.2          | 0.000        | 0.031        | -0.015        | 0.030        |
| SHG                | 95438.4          | 93266.5          | 1735764.5         | 1491112.4         | 0.000        | 0.031        | -0.014        | 0.027        |
| SKM                | 1045050.9        | 840533.4         | 226005.1          | 228119.7          | 0.000        | 0.031        | -0.016        | 0.033        |
| WF                 | 10636.5          | 20956.3          | 3620508.8         | 4277122.5         | 0.000        | 0.037        | -0.017        | 0.035        |
| <b>S. Korea</b>    | <b>512640.2</b>  | <b>411474.6</b>  | <b>1690739.6</b>  | <b>1692349.9</b>  | <b>0.000</b> | <b>0.032</b> | <b>-0.016</b> | <b>0.031</b> |
| BBVA               | 722872.8         | 1243640.4        | 32464035.7        | 40258983.1        | 0.000        | 0.026        | 0.403         | 1.398        |
| SAN                | 2039387.7        | 3916645.5        | 2689425.3         | 2315336.6         | 0.000        | 0.026        | 0.258         | 0.660        |
| TEF                | 1943805.2        | 4119102.5        | 35953083.0        | 34019971.5        | 0.000        | 0.021        | 0.403         | 1.396        |
| <b>Spain</b>       | <b>1568688.5</b> | <b>3093129.5</b> | <b>23702181.3</b> | <b>25531430.4</b> | <b>0.000</b> | <b>0.025</b> | <b>0.354</b>  | <b>1.151</b> |
| ABB                | 2216333.3        | 2356541.3        | 11355899.3        | 8806441.1         | 0.000        | 0.033        | -0.016        | 0.041        |
| CS                 | 973933.0         | 1039321.7        | 7240662.7         | 3863792.7         | 0.000        | 0.029        | -0.015        | 0.032        |
| NVS                | 1586069.1        | 1621393.9        | 6779358.6         | 3684363.5         | 0.000        | 0.014        | -0.008        | 0.015        |
| STM                | 1772087.2        | 1149713.3        | 7891434.2         | 6692322.5         | -0.001       | 0.030        | -0.017        | 0.032        |
| SYT                | 300217.0         | 303644.5         | 439709.0          | 271447.8          | 0.001        | 0.020        | -0.012        | 0.021        |
| <b>Switzerland</b> | <b>1369727.9</b> | <b>1294122.9</b> | <b>6741412.8</b>  | <b>4663673.5</b>  | <b>0.000</b> | <b>0.025</b> | <b>-0.014</b> | <b>0.028</b> |
| ASX                | 790151.1         | 887111.3         | 42810093.8        | 29386804.7        | 0.000        | 0.032        | -0.017        | 0.031        |
| AUO                | 2403933.0        | 1778126.0        | 77308646.8        | 54013052.7        | 0.000        | 0.032        | -0.018        | 0.031        |
| CHT                | 688174.6         | 737544.8         | 12347310.9        | 10330140.0        | 0.000        | 0.017        | -0.006        | 0.013        |
| SPIL               | 1403183.1        | 1265599.5        | 19086990.5        | 13175345.8        | 0.000        | 0.035        | -0.017        | 0.033        |
| TSM                | 9872584.2        | 6863761.7        | 67184614.2        | 64449361.0        | 0.000        | 0.033        | -0.011        | 0.026        |
| UMC                | 3687556.3        | 2518071.8        | 61268704.3        | 43932625.6        | -0.001       | 0.036        | -0.014        | 0.027        |
| <b>Taiwan</b>      | <b>3140930.4</b> | <b>2341702.5</b> | <b>46667726.7</b> | <b>35881221.6</b> | <b>0.000</b> | <b>0.031</b> | <b>-0.014</b> | <b>0.027</b> |
| AV                 | 74938.1          | 99254.0          | 10448589.5        | 5263321.6         | 0.000        | 0.024        | -0.014        | 0.025        |
| AZN                | 976544.6         | 1019286.7        | 4584643.6         | 3872894.5         | 0.000        | 0.018        | -0.011        | 0.021        |
| BBL                | 774249.8         | 901318.5         | 14727426.7        | 10280865.3        | 0.001        | 0.028        | -0.014        | 0.029        |
| BCS                | 1211229.8        | 2215159.1        | 40801768.7        | 35010243.8        | 0.000        | 0.033        | -0.018        | 0.038        |

|      | ADR              | Volume           | Home              | Volume            | ADR          | Return       | Home          | Return       |
|------|------------------|------------------|-------------------|-------------------|--------------|--------------|---------------|--------------|
|      | Mean             | St. Dev          | Mean              | St. Dev           | Mean         | St. Dev      | Mean          | St. Dev      |
| BP   | 4845448.2        | 9738717.8        | 46219375.9        | 37031632.6        | 0.000        | 0.018        | -0.012        | 0.021        |
| BT   | 127544.8         | 184707.8         | 31275582.1        | 28920061.2        | 0.000        | 0.024        | -0.015        | 0.027        |
| CUK  | 122466.7         | 146596.0         | 1478402.5         | 1255890.4         | 0.000        | 0.024        | -0.014        | 0.029        |
| DEO  | 457141.0         | 334142.8         | 11324875.5        | 10801784.3        | 0.000        | 0.016        | -0.011        | 0.020        |
| GSK  | 1421199.8        | 1191401.9        | 12689314.1        | 10493142.2        | 0.000        | 0.017        | -0.011        | 0.020        |
| HSBC | 1341768.5        | 1527827.5        | 38812387.8        | 25734493.6        | 0.000        | 0.018        | -0.010        | 0.021        |
| IHG  | 156098.6         | 268945.7         | 1988399.9         | 1562739.5         | 0.000        | 0.024        | -0.013        | 0.025        |
| UK   | <b>1046239.1</b> | <b>1602487.1</b> | <b>19486433.3</b> | <b>15475188.1</b> | <b>0.000</b> | <b>0.022</b> | <b>-0.013</b> | <b>0.025</b> |

Figure 1: XY plots of risk denoted by standard deviation against mean return

The row contains mean daily returns (X axis) and the corresponding standard deviations (Y axis) for home shares and ADRs in columns 1 and 2 respectively.



**Table 3: Country wise summaries of estimated parameters of the following bivariate VAR model for ADR and home security volume**

$$v_{i,t}^{us} = \alpha + \beta v_{i,t-1}^{us} + \phi v_{t-1}^{us} + V_{i,t}^{us} \quad (1a)$$

$$v_{i,t}^h = \alpha + \mu_i v_{i,t-1}^h + \theta_j v_{t-j}^h + V_{i,t}^h \quad (1b)$$

|                               | ADR                                         |                                            |              |                                           | Home                                         |              |
|-------------------------------|---------------------------------------------|--------------------------------------------|--------------|-------------------------------------------|----------------------------------------------|--------------|
| Country (# of security pairs) | Mean $\beta$ ; significant +; significant - | Mean $\phi$ ; significant +; significant - | Median $R^2$ | Mean $\mu$ ; significant +; significant - | Mean $\theta$ ; significant +; significant - | Median $R^2$ |
| Australia (6)                 | -.392; 0; 6                                 | -.017; 1; 2                                | .172         | -.357; 0; 6                               | .057; 5; 1                                   | .140         |
| Belgium (2)                   | -.366; 0; 2                                 | .025; 1; 0                                 | .130         | -.401; 0; 2                               | .053; 2; 0                                   | .140         |
| Brazil (15)                   | -.392; 0; 15                                | .027; 10; 1                                | .153         | -.403; 0; 15                              | .027; 7; 0                                   | .148         |
| Chile (10)                    | -.417; 0; 10                                | .006; 1; 0                                 | .177         | -.398; 0; 10                              | .044; 7; 0                                   | .150         |
| Denmark (1)                   | -.424; 0; 1                                 | .025; 0; 0                                 | .177         | -.379; 0; 1                               | .057; 1; 0                                   | .140         |
| Finland (1)                   | -.310; 0; 1                                 | -.023; 0; 0                                | .103         | -.419; 0; 1                               | .210; 1; 0                                   | .130         |
| France (5)                    | -.384; 0; 5                                 | .002; 1; 1                                 | .152         | -.376; 0; 5                               | .065; 5; 0                                   | .124         |
| Germany (1)                   | -.437; 0; 1                                 | .009; 0; 0                                 | .192         | -.374; 0; 1                               | .042; 1; 0                                   | .132         |
| Hong Kong (12)                | -.399; 0; 12                                | .001; 0; 2                                 | .157         | -.355; 0; 12                              | .104; 12; 0                                  | .122         |
| India (6)                     | -.373; 0; 6                                 | .013; 1; 1                                 | .131         | -.368; 0; 6                               | .147; 6; 0                                   | .154         |
| Indonesia (1)                 | -.402; 0; 1                                 | .023; 0; 0                                 | .160         | -.379; 0; 1                               | .128; 1; 0                                   | .143         |
| Ireland (3)                   | -.341; 0; 3                                 | .001; 0; 1                                 | .140         | -.413; 0; 3                               | .078; 3; 0                                   | .164         |
| Israel (2)                    | -.358; 0; 2                                 | -.042; 0; 1                                | .152         | -.422; 0; 2                               | .062; 2; 0                                   | .174         |
| Italy (3)                     | -.394; 0; 3                                 | -.005; 0; 0                                | .155         | -.362; 0; 3                               | .051; 3; 0                                   | .117         |
| Japan (17)                    | -.396; 0; 17                                | -.021; 0; 6                                | .158         | -.365; 0; 17                              | .069; 17; 0                                  | .131         |
| Mexico (12)                   | -.372; 0; 12                                | .024; 5; 0                                 | .128         | -.375; 0; 12                              | -.007; 3; 0                                  | .140         |
| Netherlands (3)               | -.375; 0; 3                                 | -.007; 0; 0                                | .140         | -.379; 0; 3                               | .080; 3; 0                                   | .127         |
| Norway (1)                    | -.403; 0; 1                                 | -.005; 0; 1                                | .169         | -.401; 0; 1                               | .083; 1; 0                                   | .155         |
| Philippines (1)               | -.389; 0; 1                                 | -.027; 0; 1                                | .153         | -.401; 0; 1                               | .132; 1; 0                                   | .170         |
| Portugal (1)                  | -.430; 0; 1                                 | .036; 1; 0                                 | .181         | -.380; 0; 1                               | .062; 1; 0                                   | .140         |
| S Africa (6)                  | -.398; 0; 6                                 | -.020; 0; 3                                | .169         | -.389; 0; 6                               | .109; 6; 0                                   | .158         |
| S Korea (7)                   | -.381; 0; 7                                 | -.029; 0; 4                                | .141         | -.386; 0; 7                               | .106; 7; 0                                   | .160         |
| Spain (3)                     | -.397; 0; 3                                 | -.011; 0; 0                                | .166         | -.391; 0; 3                               | .034; 3; 0                                   | .141         |
| Switzerland (5)               | -.366; 0; 5                                 | -.008; 1; 1                                | .135         | -.378; 0; 5                               | .082; 5; 0                                   | .123         |
| Taiwan (6)                    | -.396; 0; 6                                 | -.030; 1; 3                                | .163         | -.368; 0; 6                               | .132; 6; 0                                   | .153         |
| UK (11)                       | -.378; 0; 11                                | -.004; 1; 1                                | .137         | -.415; 0; 11                              | .076; 11; 0                                  | .157         |
| All                           |                                             |                                            |              |                                           |                                              |              |

**Table 4: Country wise summaries of estimated parameters of the following bivariate VAR model for ADR and home security return,  $r_t$**

$$r_{i,t}^{us} = a_1 + b_1 r_{i,t-1}^{us} + c_1 r_{i,t-1}^h + d_1 r_{i,t-1}^{us} V_{i,t-1}^{us} + e_1 r_{i,t-1}^h V_{i,t-1}^h + f_1 r_{i,t-1}^{us} + g_1 r_{i,t-1}^h + e_{i,t}^{us} \quad (3c)$$

$$r_{i,t}^h = a_2 + b_2 r_{i,t-1}^h + c_2 r_{i,t-1}^{us} + d_2 r_{i,t-1}^h V_{i,t-1}^h + e_2 r_{i,t-1}^{us} V_{i,t-1}^{us} + f_2 r_{i,t-1}^h + g_2 r_{i,t-1}^{us} + e_{i,t}^h \quad (3d)$$

| Country (#<br>of security<br>pairs) | Mean $b_1$ ;<br>+ sign<br>- sign | Mean $c_1$ ;<br>+ sign<br>- sign | Mean $d_1$ ;<br>+ sign<br>- sign | Mean $e_1$ ;<br>+ sign<br>- sign | Mean $b_2$ ;<br>+ sign<br>- sign | Mean $c_2$ ;<br>+ sign<br>- sign | Mean $d_2$ ;<br>+ sign<br>- sign | Mean $e_2$ ;<br>+ sign<br>- sign |
|-------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| Australia                           | -.071;                           | -.005;                           | .158;                            | .014;                            | .044;                            | -.005;                           | .232;                            | -.004;                           |
| Belgium                             | -.104;                           | -.015;                           | .071;                            | -.018;                           | .266;                            | -.041;                           | .121;                            | .320;                            |
| Brazil                              | -.043;                           | -.018;                           | .033;                            | .018;                            | .271;                            | .025;                            | .019;                            | -.038;                           |
| Chile                               | -.046;                           | -.008;                           | .035;                            | -.006;                           | .297;                            | -.005;                           | .157;                            | -.022;                           |
| Denmark                             | -.075;                           | .034;                            | -.026;                           | .057;                            | .134;                            | .126;                            | .195;                            | .002;                            |
| Finland                             | -.040;                           | -.003;                           | .0002;                           | .002;                            | .990;                            | .009;                            | -.001;                           | -.145;                           |
| France                              | -.052;                           | .013;                            | .012;                            | -.001;                           | .710;                            | -.017;                           | -.110;                           | -.065;                           |
| Germany                             | -.091;                           | .005;                            | .066;                            | .015;                            | .262;                            | .009;                            | .165;                            | -.089;                           |
| Hong Kong                           | -.125;                           | -.007;                           | .038;                            | .027;                            | .125;                            | -.077;                           | .313;                            | -.033;                           |
| India                               | -.077;                           | .046;                            | -.011;                           | -.007;                           | .194;                            | -.028;                           | .213;                            | .050;                            |
| Indonesia                           | -.060;                           | -.049;                           | .051;                            | .001;                            | .105;                            | -.052;                           | .316;                            | -.055;                           |
| Ireland                             | -.064;                           | .031;                            | .040;                            | -.014;                           | .208;                            | -.067;                           | .325;                            | -.003;                           |
| Israel                              | -.042;                           | -.0004;                          | .087;                            | -.015;                           | .061                             | -.090;                           | .305;                            | .105;                            |
| Italy                               | -.072;                           | -.020;                           | -.003;                           | -.010;                           | .165;                            | -.001;                           | .235;                            | -.025;                           |
| Japan                               | -.099;                           | .003;                            | .039;                            | .038;                            | .075;                            | -.045;                           | .332;                            | .009;                            |
| Mexico                              | -.025;                           | .017;                            | .041;                            | -.008;                           | .234;                            | .010;                            | .107;                            | -.005;                           |
| Netherlands                         | -.031;                           | .001;                            | -.003;                           | .002;                            | .980;                            | -.010;                           | -.283;                           | -.117;                           |
| Norway                              | -.089;                           | -.006;                           | .005;                            | -.046;                           | .054;                            | .019;                            | .309;                            | -.053;                           |
| Philippines                         | -.007;                           | -.0004;                          | -.009;                           | .021;                            | -.002;                           | -.070;                           | .507;                            | .027;                            |
| Portugal                            | -.047;                           | -.020;                           | -.014;                           | .001;                            | .097;                            | -.065;                           | .308;                            | .014;                            |
| S Africa                            | -.047;                           | -.046;                           | .019;                            | .012;                            | .232;                            | -.014;                           | .274;                            | -.028;                           |
| S Korea                             | -.069;                           | .022;                            | -.020;                           | .072;                            | .151;                            | .005;                            | .254;                            | .007;                            |
| Spain                               | -.034;                           | -.005;                           | -.001;                           | -.001;                           | .968;                            | .017;                            | -.469;                           | .327;                            |
| Switzerland                         | -.076;                           | .035;                            | -.006;                           | .051;                            | .135;                            | -.002;                           | .286;                            | .077;                            |
| Taiwan                              | -.113;                           | .007;                            | .039;                            | -.0003;                          | .148;                            | -.004;                           | .228;                            | -.019;                           |
| UK                                  | -.072;                           | .001;                            | .013;                            | .028;                            | .163;                            | -.004;                           | .276;                            | -.032;                           |
| All-Model3                          | -.069;4,108                      | .009;23,23                       | .025;64,16                       | .013;43,16                       | .232;129,5                       | -.015;24;57                      | .19;124,11                       | -.004;25,51                      |
| All-Model11                         | -.046;8,70                       | .005;23,25                       | .033;55,11                       | .025;50,8                        | .235;111,4                       | -.012;30,38                      | .244;109,8                       | -.012;18,43                      |
| All-Model2                          | -.046;9,65                       | .004;8,19                        | .042;58,4                        | .032;33,4                        | .194;95,5                        | -.011;4,77                       | .289;100,1                       | -.156;3,89                       |
| All-Model4                          | -.068;4,111                      | .004;26,29                       | .032;70,13                       | -3e <sup>-5</sup> ;35,31         | .238;132,4                       | -.038;12,76                      | .19;124,11                       | .001;24,22                       |



**Table 5: Conditional Expected Absolute Value of Volume Induced Return Spillover Parameter**

$$Y_i = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1$$

$X_1 = \text{Ln (MVE)}$

$X_2 = \text{Mktdummy} \{1 \text{ if emerging; } 0, \text{ if developed}\}$

$X_3 = \text{Condummy} \{1, \text{ if Asia-Pacific; } -1, \text{ if Europe; } 0, \text{ if Americas}\}$

$X_4 = \text{Timeolap} \{1, \text{ if overlap; } 0, \text{ if no overlap}\}$

$X_5 = \text{Lawdummy} \{1, \text{ if English common law; } 0, \text{ if Civil law}\}$

$X_6 = \text{Parasigndummy} \{1, \text{ if } + \text{ significant; } -1, \text{ if } - \text{ significant; } 0, \text{ if not significant}\}$

|                       | ADR       | ADR       | Home       | Home      |
|-----------------------|-----------|-----------|------------|-----------|
|                       | Y = D1    | Y = E1    | Y= D2      | Y = E2    |
| $E(Y X_2=0, X_6=0)$   | .022*     | .019*     | .050*      | .057*     |
| $E(Y X_2=1, X_6=0)$   | + .012*** | + 0       | + 0        | + 0       |
| $E(Y X_2=0, X_6=1)$   | + .060*   | + .057*   | + 0        | + .088*   |
| $E(Y X_2=1, X_6=1)$   | + .072*   | + .057*   | + 0        | + .088*   |
| $E(Y X_2=0, X_6=-1)$  | + -.060*  | + -.057*  | + 0        | + -.088*  |
| $E(Y X_2=1, X_6=-1)$  | + -.048*  | + -.057*  | + 0        | + -.088*  |
|                       |           |           |            |           |
| $E(Y X_3=0, X_6=0)$   | .028*     | .021*     | .045*      | .048*     |
| $E(Y X_3=1, X_6=0)$   | + 0       | + .007**  | + 0        | + -.024** |
| $E(Y X_3=-1, X_6=0)$  | + 0       | + -.007** | + 0        | + .024**  |
| $E(Y X_3=0, X_6=1)$   | + .059*   | + .055*   | + 0        | + .093*   |
| $E(Y X_3=1, X_6=1)$   | + .059*   | + .062*   | + 0        | + .069*   |
| $E(Y X_3=-1, X_6=1)$  | + .059*   | + .048*   | + 0        | + .117*   |
| $E(Y X_3=0, X_6=-1)$  | + -.059*  | + -.055*  | + 0        | + -.093*  |
| $E(Y X_3=1, X_6=-1)$  | + -.059*  | + -.048*  | + 0        | + -.117*  |
| $E(Y X_3=-1, X_6=-1)$ | + -.059*  | + -.062*  | + 0        | + -.069*  |
|                       |           |           |            |           |
| $E(Y X_4=0, X_6=0)$   | .022*     | .030*     | .059*      | .030**    |
| $E(Y X_4=1, X_6=0)$   | + 0       | + -.013** | + -.024*   | + 0       |
| $E(Y X_4=0, X_6=1)$   | + .061*   | + .054*   | + .019***  | + .094*   |
| $E(Y X_4=1, X_6=1)$   | + .061*   | + .041*   | + -.005*   | + .094*   |
| $E(Y X_4=0, X_6=-1)$  | + -.061*  | + -.054*  | + -.019*** | + -.094*  |
| $E(Y X_4=1, X_6=-1)$  | + -.061*  | + -.067*  | + -.043*   | + -.094*  |
|                       |           |           |            |           |
| $E(Y X_5=0, X_6=0)$   | .029*     | .021*     | .050*      | .045*     |
| $E(Y X_5=1, X_6=0)$   | + 0       | + 0       | + .013*    | + 0       |
| $E(Y X_5=0, X_6=1)$   | + .059*   | + .057*   | + .018***  | + .092*   |
| $E(Y X_5=1, X_6=1)$   | + .059*   | + .057*   | + .031*    | + .092*   |
| $E(Y X_5=0, X_6=-1)$  | + -.059*  | + -.057*  | + -.018*** | + -.092*  |
| $E(Y X_5=1, X_6=-1)$  | + -.059*  | + -.057*  | + -.005*   | + -.092*  |

Table 6A: Country wise summary of testing the following hypotheses Granger causality via Wald test; number of rejections as per significant Chi-square statistics are reported.

- ADR (home) security lag volatility Granger cause home (ADR) volatility

$$vol_t^{us} = \alpha_2 + \sum_{m=1}^j \beta_{2m} V_{t-m}^{us} + \sum_{n=1}^j \theta_{2n} vol_{t-n}^{us} + \varepsilon_t^{us}$$

$$vol_t^h = \alpha_2 + \sum_{m=1}^j \beta_{2m} V_{t-m}^h + \sum_{n=1}^j \theta_{2n} vol_{t-n}^h + \varepsilon_t^h$$

|                               | ADR        |                 | Home           |                 | Granger causality                                                                        |                                                                                           |
|-------------------------------|------------|-----------------|----------------|-----------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Country (# of security pairs) | $B_{1m}=0$ | $\theta_{1n}=0$ | $\beta_{2m}=0$ | $\theta_{2n}=0$ | $H_0: \beta_{2m}=0$<br>Home share $\sigma^2 \rightarrow$<br>$\rightarrow$ ADR $\sigma^2$ | $H_0: \theta_{1n}=0$<br>ADR $\sigma^2 \rightarrow$<br>$\rightarrow$ Home share $\sigma^2$ |
| Australia (6)                 | .082;      | .133;           | .232;          | .076;           | 6                                                                                        | 6                                                                                         |
| Belgium (2)                   | .035;      | .133;           | .194;          | .112;           | 2                                                                                        | 2                                                                                         |
| Brazil (15)                   | .092;      | .158;           | .156;          | .090;           | 15                                                                                       | 15                                                                                        |
| Chile (10)                    | .150;      | .160;           | .211;          | .068;           | 10                                                                                       | 10                                                                                        |
| Denmark (1)                   | .002;      | .003;           | .181;          | .200;           | 1                                                                                        | 1                                                                                         |
| Finland (1)                   | .042;      | .000;           | .002;          | .378;           | 0                                                                                        | 1                                                                                         |
| France (5)                    | .099;      | .068;           | .134;          | .118;           | 2                                                                                        | 2                                                                                         |
| Germany (1)                   | .001;      | .176;           | .302;          | .003;           | 1                                                                                        | 1                                                                                         |
| Hong Kong (12)                | .128;      | .118;           | .111;          | .197;           | 12                                                                                       | 12                                                                                        |
| India (6)                     | .121;      | .046;           | .195;          | .063;           | 6                                                                                        | 6                                                                                         |
| Indonesia (1)                 | .166;      | .105;           | .166;          | .055;           | 1                                                                                        | 1                                                                                         |
| Ireland (3)                   | .060;      | .071;           | .075;          | .123;           | 2                                                                                        | 3                                                                                         |
| Israel (2)                    | .033;      | .025;           | .190;          | .007;           | 0                                                                                        | 2                                                                                         |
| Italy (3)                     | .062;      | .186;           | .242;          | .031;           | 3                                                                                        | 2                                                                                         |
| Japan (17)                    | .099;      | .165;           | .178;          | .165;           | 16                                                                                       | 17                                                                                        |
| Mexico (12)                   | .013;      | .135;           | .115;          | .119;           | 9                                                                                        | 8                                                                                         |
| Netherlands (3)               | .155;      | .011;           | .021;          | .328;           | 1                                                                                        | 3                                                                                         |
| Norway (1)                    | .120;      | .181;           | .144;          | .081;           | 1                                                                                        | 1                                                                                         |
| Philippines (1)               | .066;      | .071;           | .270;          | .112;           | 1                                                                                        | 1                                                                                         |
| Portugal (1)                  | .081;      | .098;           | .106;          | .131;           | 1                                                                                        | 1                                                                                         |
| S Africa (6)                  | .153;      | .081;           | .228;          | .068;           | 5                                                                                        | 5                                                                                         |
| S Korea (7)                   | .154;      | .222;           | .134;          | .165            | 7                                                                                        | 7                                                                                         |
| Spain (3)                     | .193;      | -.0002;         | .403;          | -2.33           | 2                                                                                        | 0                                                                                         |
| Switzerland (5)               | .101;      | .180;           | .147;          | .295;           | 5                                                                                        | 5                                                                                         |
| Taiwan (6)                    | .104;      | .157;           | .136;          | .045;           | 6                                                                                        | 6                                                                                         |
| UK (11)                       | -.039;     | .194;           | .231;          | .013;           | 9                                                                                        | 10                                                                                        |

Table 6B: Country wise summary of testing the following hypotheses Granger causality via Wald test; Chi-square statistics are reported.

- ADR (home) security lag volatility Granger cause home (ADR) volatility

- ADR: Volatility (volume) Granger cause Volume (Volatility)

- Home: Volatility (volume) Granger cause Volume (Volatility)

$$\sigma_t^2(A) = a_0 + a_1\sigma_{t-1}^2(A) + a_2V_{t-1}^A + a_3\sigma_{t-1}^2(H) + a_4V_{t-1}^H + \varepsilon_1 \quad (5a)$$

$$\sigma_t^2(H) = a_0 + a_1\sigma_{t-1}^2(A) + a_2V_{t-1}^A + a_3\sigma_{t-1}^2(H) + a_4V_{t-1}^H + \varepsilon_1 \quad (5b)$$

|             | ADR                |           |                     |           | Home               |           |                     |           |
|-------------|--------------------|-----------|---------------------|-----------|--------------------|-----------|---------------------|-----------|
| Country     | $\sigma^2$ -A(t-1) | V -A(t-1) | $\sigma^2$ -H (t-1) | V -H(t-1) | $\sigma^2$ -H(t-1) | V -H(t-1) | $\sigma^2$ -A (t-1) | V -A(t-1) |
| Australia   | .076;              | .001;     | .318;               | .000;     | .237;              | .000;     | .073;               | .000;     |
| Belgium     | .025;              | .000;     | .145;               | .000;     | .191;              | .000;     | .110;               | .000      |
| Brazil      | .092;              | .000;     | .163;               | .000;     | .163;              | .000;     | .086;               | .000;     |
| Chile       | .150;              | .000;     | .160;               | .000;     | .222;              | .000;     | .066;               | .000;     |
| Denmark     | .002;              | .000;     | .003;               | .000;     | .182;              | .000;     | .201;               | .000;     |
| Finland     | .043;              | .000;     | -.0001;             | .000;     | .002;              | -.005;    | .376;               | .000;     |
| France      | .098;              | .000;     | .070;               | .000;     | .136;              | .000;     | .057;               | .000;     |
| Germany     | -.078;             | .000;     | .177;               | .000;     | .299;              | .000;     | -.013;              | .000;     |
| Hong Kong   | .130;              | .000;     | .116;               | .000;     | .118;              | .000;     | .197;               | .000;     |
| India       | .075;              | .000;     | .078;               | .000;     | .242;              | .000;     | -.007;              | .000;     |
| Indonesia   | .164;              | .000;     | .107;               | .000;     | .167;              | .000;     | .054;               | .000;     |
| Ireland     | .067;              | .000;     | .074;               | .000;     | .078;              | .000;     | .130;               | .000;     |
| Israel      | .038;              | .000;     | .031;               | .000;     | .211;              | .000;     | .008;               | .000;     |
| Italy       | .067;              | .000;     | .183;               | .000;     | .233;              | .000;     | .036;               | .000;     |
| Japan       | .100;              | .000;     | .168;               | .000;     | .171;              | .000;     | .171;               | .000;     |
| Mexico      | .002;              | .000;     | .139;               | .000;     | .124;              | .000;     | .124;               | .000;     |
| Netherlands | .156;              | .000;     | .010;               | .000;     | .020;              | .000;     | .324;               | .000;     |
| Norway      | .118;              | .000;     | .189;               | .000;     | .143;              | .000;     | .080;               | .000;     |
| Philippines | .067;              | .000;     | .070;               | .000;     | .276;              | .000;     | .120;               | .000;     |
| Portugal    | .082;              | .000;     | .099;               | .000;     | .106;              | .000;     | .131;               | .000;     |
| S Africa    | .159;              | .000;     | .082;               | .000;     | .226;              | .000;     | .074;               | .000;     |
| S Korea     | .153;              | .000;     | .232;               | .000;     | .138;              | .000;     | .161;               | .000;     |
| Spain       | .194;              | .000;     | -.0001;             | .000;     | .404;              | -.050;    | -.973;              | .000;     |
| Switzerland | .097;              | .000;     | .191;               | .000;     | .135;              | .000;     | .310;               | .000;     |
| Taiwan      | .109;              | .000;     | .158;               | .000;     | .135;              | .000;     | .046;               | .000;     |
| UK          | -.049;             | .000;     | .205;               | .000;     | .242;              | .000;     | .008;               | .000;     |
| All         | .089;              | .000;     | .146;               | .000;     | .175;              | -.001;    | .089;               | .000;     |

Table 7: Country wise summary of volume effect on volatility spillover for US and home securities

$$\sigma_{i,t}^{us} = \alpha_1 + \beta_1 \sigma_{i,t-1}^{us} + \theta_1 \sigma_{i,t-1}^h + \gamma_1 \sigma_{i,t-1}^{us} V_{t-1}^{us} + \pi_{1q} \sigma_{i,t-n}^h V_{i,t-q}^h + \eta_{i,t}^{us} \quad (5a)$$

$$\sigma_{i,t}^h = \alpha_2 + \beta_2 \sigma_{i,t-1}^{us} + \theta_2 \sigma_{i,t-1}^h + \lambda_2 \sigma_{i,t-1}^{us} V_{i,t-1}^{us} + \phi_2 \sigma_{i,t-1}^h V_{i,t-1}^h + \eta_t^h \quad (5b)$$

| Country<br>(Nos. of<br>security<br>pairs) |                                              |                                           |                                             |                                              | Home                                      |                                             |
|-------------------------------------------|----------------------------------------------|-------------------------------------------|---------------------------------------------|----------------------------------------------|-------------------------------------------|---------------------------------------------|
|                                           | Mean $\gamma_1$ ;<br>% +/-;<br>% significant | Mean $\pi_1$ ;<br>% +/-;<br>% significant | High R <sup>2</sup> ;<br>Low R <sup>2</sup> | Mean $\gamma_2$ ;<br>% +/-;<br>% significant | Mean $\pi_2$ ;<br>% +/-;<br>% significant | High R <sup>2</sup> ;<br>Low R <sup>2</sup> |
| Australia (6)                             | -.017;                                       | -.029;                                    | .087;                                       | .057;                                        | -.024;                                    | .125;                                       |
| Belgium (2)                               | .026;                                        | -.049;                                    | .034;                                       | -.125;                                       | .031;                                     | .098;                                       |
| Brazil (15)                               | .003;                                        | -.100;                                    | .056;                                       | .009;                                        | -.027;                                    | .068;                                       |
| Chile (10)                                | -.049;                                       | -.029;                                    | .085;                                       | .006;                                        | -.035;                                    | .081;                                       |
| Denmark (1)                               | -.036;                                       | -.001;                                    | .007;                                       | .112;                                        | -.081;                                    | .128;                                       |
| Finland (1)                               | -.032;                                       | -.007;                                    | .004;                                       | -.041;                                       | .056;                                     | -.001;                                      |
| France (5)                                | -.024;                                       | -.047;                                    | .038;                                       | -.035;                                       | .044;                                     | .058;                                       |
| Germany (1)                               | -.028;                                       | -.046;                                    | .018;                                       | -.075;                                       | -.038;                                    | .095;                                       |
| Hong Kong (12)                            | -.073;                                       | -.120;                                    | .084;                                       | -.051;                                       | -.053;                                    | .068;                                       |
| India (6)                                 | -1.00;                                       | .026;                                     | .034;                                       | -.011;                                       | -.121;                                    | .082;                                       |
| Indonesia (1)                             | -.068;                                       | -.179;                                    | .064;                                       | -.027;                                       | -.113;                                    | .066;                                       |
| Ireland (3)                               | -.111;                                       | -.078;                                    | .041;                                       | -.067;                                       | -.175;                                    | .060;                                       |
| Israel (2)                                | -.035;                                       | -.025;                                    | .007;                                       | .223;                                        | -.166;                                    | .103;                                       |
| Italy (3)                                 | .005;                                        | .076;                                     | .068;                                       | .369;                                        | .008;                                     | .085;                                       |
| Japan (17)                                | -.016;                                       | -.067;                                    | .079;                                       | .035;                                        | -.027;                                    | .101;                                       |
| Mexico (12)                               | -.061;                                       | .002;                                     | .046;                                       | -.0001;                                      | .007;                                     | .080;                                       |
| Netherlands (3)                           | -.019;                                       | -.076;                                    | .033;                                       | -.130;                                       | .067;                                     | .007;                                       |
| Norway (1)                                | .099;                                        | -.018;                                    | .065;                                       | -.064;                                       | .002;                                     | .050;                                       |
| Philippines (1)                           | -.058;                                       | -.088;                                    | .016;                                       | .183;                                        | .003;                                     | .136;                                       |
| Portugal (1)                              | .04;                                         | -.044;                                    | .028;                                       | .135;                                        | .147;                                     | .063;                                       |
| S Africa (6)                              | -.018;                                       | -.003;                                    | .048;                                       | .025;                                        | .015;                                     | .107;                                       |
| S Korea (7)                               | -.056;                                       | -.045;                                    | .088;                                       | .056;                                        | -.032;                                    | .110;                                       |
| Spain (3)                                 | -.042;                                       | .001;                                     | .038;                                       | -.565;                                       | -15.8;                                    | .216                                        |
| Switzerland (5)                           | -.025;                                       | .045;                                     | .116;                                       | -.020;                                       | .149;                                     | .227;                                       |
| Taiwan (6)                                | -.048;                                       | .066;                                     | .030;                                       | .035;                                        | -.004;                                    | .040;                                       |
| UK (11)                                   | -.078;                                       | -.028;                                    | .069;                                       | -.029;                                       | -.074;                                    | .085                                        |
| All                                       | -0.038;30,111                                | -0.04;26,115                              | 0.06                                        | -0.08;49,92                                  | -0.36;31,110                              | 0.09                                        |

**Table 8: Conditional Expected Absolute Value of Volume Induced Volatility Spillover Parameter**

$$Y_i = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1$$

$X_1 = \text{Ln (MVE)}$

$X_2 = \text{Mktdummy}$  {1 if emerging; 0, if developed}

$X_3 = \text{Condummy}$  {1, if Asia-Pacific; -1, if Europe; 0, if Americas}

$X_4 = \text{Timeolap}$  {1, if overlap; 0, if no overlap}

$X_5 = \text{Lawdummy}$  {1, if English common law; 0, if Civil law}

$X_6 = \text{Parasigndummy}$  {1, if + significant; -1, 0, if – significant}

|                      | ADR      | ADR          | Home       | Home       |
|----------------------|----------|--------------|------------|------------|
|                      | Y = D1   | Y = E1       | Y = D2     | Y = E2     |
| $E(Y X_2=0, X_6=0)$  | 0        | .0001*       | .003**     | .001*      |
| $E(Y X_2=1, X_6=0)$  | + 0      | + 0          | + -.003*** | + -.001**  |
| $E(Y X_2=0, X_6=1)$  | + .001*  | + .0001*     | + 0        | + 0        |
| $E(Y X_2=1, X_6=1)$  | + .001*  | + .0001*     | + -.003*** | + -.001**  |
|                      |          |              |            |            |
| $E(Y X_3=0, X_6=0)$  | 0        | .0001*       | .002***    | .0004**    |
| $E(Y X_3=1, X_6=0)$  | + 0      | + 0          | + -.002*** | + -.0005** |
| $E(Y X_3=-1, X_6=0)$ | + 0      | + 0          | + .002***  | + .0005**  |
| $E(Y X_3=0, X_6=1)$  | + .001*  | + .0002*     | + 0        | + 0        |
| $E(Y X_3=1, X_6=1)$  | + .001*  | + .0002*     | + -.002*** | + -.0005   |
| $E(Y X_3=-1, X_6=1)$ | + .001*  | + .0002*     | + .002***  | + .0005    |
|                      |          |              |            |            |
| $E(Y X_4=0, X_6=0)$  | 0        | .0001*       | 0          | 0          |
| $E(Y X_4=1, X_6=0)$  | + 0      | + -.00001*** | + 0        | + 0        |
| $E(Y X_4=0, X_6=1)$  | + .116*  | + .002*      | + 0        | + 0        |
| $E(Y X_4=1, X_6=1)$  | + .116*  | + .002*      | + 0        | + 0        |
|                      |          |              |            |            |
| $E(Y X_5=0, X_6=0)$  | .0001*** | .0001*       | 0          | .0003***   |
| $E(Y X_5=1, X_6=0)$  | + 0      | + .00002***  | + 0        | + 0        |
| $E(Y X_5=0, X_6=1)$  | + .001*  | + .001*      | + 0        | + 0        |
| $E(Y X_5=1, X_6=1)$  | + .001*  | + .00012*    | + 0        | + 0        |

**Appendix A: Sample description of 170 Developed and Emerging Markets ADRs listed on NYSE system and their underlying shares listed in the respective home countries**

| <b>Ticker</b> | <b>Name</b>                 | <b>Primary<br/>Exchange<br/>Country</b> | <b>Industry</b> | <b>1 ADR=<br/>X shares</b> | <b>Date of<br/>ADR<br/>[primary]<br/>listing</b> |
|---------------|-----------------------------|-----------------------------------------|-----------------|----------------------------|--------------------------------------------------|
| AWCMY (AWC)*  | Alumina Ltd                 | Australia                               | Metal & Mining  | 4                          | 1/90                                             |
| BHP (BHP)     | BHP Billiton                | Australia                               | Metal & Mining  | 2                          | 5/87                                             |
| JHX (JHX)     | James Hardie Industries     | Australia                               | Bldg. Mat.      | 1                          | 10/01                                            |
| OBTLY (OEC)   | Orbital Corp Ltd            | Australia                               | Engineer        | .16                        | 8/11                                             |
| SSN (SSN)     | Samson Oil and Gas          | Australia                               | Oil & Gas       | 2                          | 1/08                                             |
| SMSMY (SGM)*  | Sims Group Ltd              | Australia                               | IT              | 1                          | 3/08                                             |
| WBK (WBC)     | Westpac Banking Corp        | Australia                               | Fin Serv.       | 1                          | 3/89                                             |
| BUD (ABI)     | Anheuser Busch              | Belgium                                 | Food & Drugs    | 1                          | 9/09                                             |
| DEG (DELB)    | Delhaize Group              | Belgium                                 | Retail          | .25                        | 4/01                                             |
| NVO (NOVOB)   | Novo-Nordisk                | Denmark                                 | Food & Drugs    | 1                          | 7/81                                             |
| NOK (NOK1V)   | Nokia Corp                  | Finland                                 | Comm. Equip     | 1                          | 7/94                                             |
| ALU (ALU)     | Alcatel-Lucent              | France                                  | Comm. Equip     | 1                          | 5/92                                             |
| CGG (CGG)     | CGG                         | France                                  | Oil & Gas       | 1                          | 5/97                                             |
| ORAN (ORA)    | Oran GE SA (France Telecom) | France                                  | Telecom         | 1                          | 10/97                                            |
| SNY (SAN)     | Sanofi SA                   | France                                  | Food & Drugs    | .5                         | 7/02                                             |
| TOT (FP)      | Total SA                    | France                                  | Oil & Gas       | 1                          | 10/91                                            |
| VE (VIE)*     | Veolia Environment          | France                                  |                 | 1                          | 10/01                                            |
| FMS (FME)     | Fresenius AG                | Germany                                 | Healthcare      | .5                         | 9/96                                             |
| SAP (SAP)     | SAP SE                      | Germany                                 | Software        | 1                          | 8/98                                             |
| SI (SIEGY)*   | Siemens AG                  | Germany                                 | Mfg.            | 1                          | 3/01                                             |
| CRH (CRH)     | CRH PLC                     | Ireland                                 | Bldg. Mat.      | 1                          | 3/06                                             |
| ELN (ELN)     | Elan Corp PLC               | Ireland                                 |                 | 1                          | 1/95                                             |
| IRE (BKIR)    | Bank of Ireland             | Ireland                                 | Fin Serv.       | 40                         | 9/96                                             |
| BSI (BSI)     | Blue-Square                 | Israel                                  | Retail          | 10                         | 8/96                                             |
| TEVA (TEVA)   | Teva Pharmaceuticals        | Israel                                  | Food & Drugs    | 1                          | 5/12                                             |
| E (ENI)       | ENI SpA                     | Italy                                   | Oil & Gas       | 2                          | 11/95                                            |
| LUX (LUX)     | Luxotica Group SpA          | Italy                                   | Consumer Goods  | 1                          | 1/90                                             |

| <b>Ticker</b> | <b>Name</b>                  | <b>Primary<br/>Exchange<br/>Country</b> | <b>Industry</b>         | <b>1 ADR=<br/>X shares</b> | <b>Date of<br/>ADR<br/>[primary]<br/>listing</b> |
|---------------|------------------------------|-----------------------------------------|-------------------------|----------------------------|--------------------------------------------------|
| TI (TIT)      | Telephone Italia             | Italy                                   | Telecom                 | 10                         | 7/95                                             |
| ATEYY (6857)* | Advantest Corp               | Japan                                   |                         | 1                          | 9/01                                             |
| CAJ (7751)    | Canon Inc                    | Japan                                   | Cap. Mfg.               | 1                          | 9/00                                             |
| HMC (7267)    | Honda Motor Corp             | Japan                                   | Cap. Mfg.               | 1                          | 2/77                                             |
| KNMCY (9766)* | Konami Corp                  | Japan                                   |                         | 1                          | 9/02                                             |
| KUBTY (6326)* | Kubota Corp                  | Japan                                   | Cap. Mfg.               | 5                          | 11/76                                            |
| KYO (6971)    | Kyocera Corp                 | Japan                                   | Cap. Mfg.               | 1                          | 5/80                                             |
| MTU (8306)    | Mitsubishi Holdings          | Japan                                   | Diversified<br>Holdings | 1                          | 9/89                                             |
| MFG (8411)    | Mizuho Financial Grp         | Japan                                   | Fin Serv.               | 2                          | 11/06                                            |
| NJ (6594)     | Nidec Corp                   | Japan                                   |                         | .25                        | 9/01                                             |
| NTT (9432)    | Nippon Tel                   | Japan                                   | Telecom                 | 1                          | 9/94                                             |
| NMR (8604)    | Nomura Holdings              | Japan                                   | Diversified<br>Holdings | 1                          | 12/01                                            |
| DCM (9437)    | NTT DoCoMo                   | Japan                                   |                         | 1                          | 3/02                                             |
| IX (8591)     | Orix Corp                    | Japan                                   |                         | 5                          | 9/98                                             |
| PCRFY (6752)* | Panasonic Corp               | Japan                                   | Cap. Mfg.               | 1                          | 12/71                                            |
| SNE (6758)    | Sony Corp                    | Japan                                   | Cap. Mfg.               | 1                          | 9/70                                             |
| SMFG (8316)   | Sumitomo Mitsui Fin Group    | Japan                                   | Diversified<br>Holdings | .2                         | 11/10                                            |
| TM (7203)     | Toyota Motors                | Japan                                   | Cap. Mfg.               | 2                          | 9/99                                             |
| MT (MT)       | Arcelor Mittal (NY Registry) | Luxembourg                              | Metal &<br>Mining       | 1                          | 8/97                                             |
| TS (TEN)      | Teneris                      | Luxembourg                              | Metal &<br>Mining       | 2                          | 12/02                                            |
| AEG (AGN)     | AEGON (NY Registry)          | Netherlands                             |                         | 1                          | 11/91                                            |
| ING (INGA)    | ING                          | Netherlands                             | Fin Serv.               | 1                          | 6/97                                             |
| PHG (PHIA)    | Phillips Electronics         | Netherlands                             | Mfg.                    | 1                          | 4/87                                             |
| ENL (REN)     | Reed Elsevier                | Netherlands                             | Publisher               | 1                          | 10/94                                            |
| UN (UNA)      | Unilever (NY Registry)       | Netherlands                             | Consumer<br>Goods       | 1                          | 12/61                                            |
| STO (STL)     | Statoil                      | Norway                                  | Oil & Gas               | 1                          | 6/01                                             |
| PT (PTC)      | Portugal Telecom             | Portugal                                | Telecom                 | 1                          | 6/95                                             |
| BBVA (BBVA)   | Banco Bilbao S.A.            | Spain                                   | Fin Serv.               | 1                          | 12/88                                            |
| SAN (SAN)     | Banco Santander S.A.         | Spain                                   | Fin Serv.               | 1                          | 7/87                                             |
| TEF (TEF)     | Tefonica S.A.                | Spain                                   | Telecom                 | 1                          | 6/87                                             |
| ABB (ABBN)    | ABB                          | Switzerland                             | Cap. Mfg.               | 1                          | 4/01                                             |
| CS (CSGN)     | Credit Suisse                | Switzerland                             | Fin Serv.               | 1                          | 9/01                                             |
| NVS (NOVN)    | Novartis AG                  | Switzerland                             |                         | 1                          | 5/00                                             |

| <b>Ticker</b> | <b>Name</b>                        | <b>Primary<br/>Exchange<br/>Country</b> | <b>Industry</b>      | <b>1 ADR=<br/>X shares</b> | <b>Date of<br/>ADR<br/>[primary]<br/>listing</b> |
|---------------|------------------------------------|-----------------------------------------|----------------------|----------------------------|--------------------------------------------------|
| STM (STM)     | STMicroelectronics<br>(NYRegistry) | Switzerland                             |                      | 1                          | 12/94                                            |
| SYT (SYNN)    | Syngenta AG                        | Switzerland                             |                      | .2                         | 11/00                                            |
| AZN (AZN)     | Astrazeneca PLC                    | UK                                      | Pharm.               | .5                         | 5/93                                             |
| AV (AV)       | Aviva PLC                          | UK                                      |                      | 2                          | 10/09                                            |
| BCS (BARC)    | Barclays                           | UK                                      | Fin Serv.            | 4                          | 9/86                                             |
| BBL (BLT)     | BHP Billiton                       | UK                                      |                      | 2                          | 6/03                                             |
| BP (BP)       | British Petroleum                  | UK                                      | Oil & Gas            | 6                          | 3/70                                             |
| BTI (BATS)    | British American Tobacco           | UK                                      | Consumer<br>Goods    | 2                          | 3/28                                             |
| BT (BT/A)     | BT Group                           | UK                                      | Telecom              | 5                          | 12/84                                            |
| CUK (CCL)     | Carnival PLC                       | UK                                      | Leisure              | 1                          | 10/00                                            |
| DEO (DGE)     | Diageo PLC                         | UK                                      |                      | 4                          | 12/97                                            |
| GSK (GSK)     | GlaxoSmithKline                    | UK                                      | Pharm.               | 2                          | 6/87                                             |
| HBC (HSBA)    | HSBC Holdings                      | UK                                      | Fin. Serv.           | 5                          | 7/99                                             |
| IHG (IHG)     | Intercontinental Hotels            | UK                                      | Leisure              | 1                          | 3/03                                             |
| LYG (LLOY)    | Lloyds Banking                     | UK                                      | Fin. Serv.           | 4                          | 11/01                                            |
| NGG (NG)      | National Grid                      | UK                                      |                      | 5                          | 10/99                                            |
| PSO (PSON)    | Pearson plc                        | UK                                      | Publisher            | 1                          | 9/00                                             |
| PUK (PRU)     | Prudential Insurance               | UK                                      | Fin Serv.            | 2                          | 6/00                                             |
| RUK (REL)     | Reed Elsevier plc                  | UK                                      | Publisher            | 1                          | 10/94                                            |
| RIO (RIO)     | Rio Tinto plc                      | UK                                      |                      | 1                          | 6/90                                             |
| SNN (SN)      | Smith and Nephew plc               | UK                                      |                      | 2                          | 11/99                                            |
| RBS (RBS)     | Royal Bank of Scotland plc         | UK                                      | Fin Serv.            | 2                          | 10/07                                            |
| UL (ULVR)     | Unilever                           | UK                                      | Consumer<br>Goods    | 1                          | 12/61                                            |
| BMA (BMA)     | Banco Marco                        | Argentina                               | Fin Serv.            | 1                          | 3/06                                             |
| BFR (FRAN)    | BBVA Banco Francais                | Argentina                               | Fin Serv.            | 3                          | 11/93                                            |
| EDN (EDN)     | Empress DC Norte                   | Argentina                               | Power &<br>Utilities | 20                         | 4/07                                             |
| IRS (IRSA)    | IRSA (Global share)                | Argentina                               | Fin Serv             | 10                         | 12/94                                            |
| NTL (PFD)     | Nortel                             | Argentina                               | Comm                 | .05                        | 6/97                                             |
| PAM (PAMP)    | Pampa Energie                      | Argentina                               | Power &<br>Utilities | 25                         | 10/09                                            |
| PZE (PESA)    | Petrobras AR                       | Argentina                               | Oil & Gas            | 1                          | 1/00                                             |
| TEO (TECOZ)   | Telecom AR                         | Argentina                               | Comm                 | 5                          | 12/94                                            |
| TGS (TGSUZ)   | Transportadora                     | Argentina                               | Oil & Gas            | 5                          | 11:94                                            |
| YPF (YPFD)    | YPF                                | Argentina                               | Oil & Gas            | 5                          | 6/93                                             |
| BBD [BBDC3]   | Banco Bradesco S.A                 | Brazil                                  | Fin Serv.            | 1                          | 11/01                                            |



| <b>Ticker</b>                 | <b>Name</b>                    | <b>Primary<br/>Exchange<br/>Country</b> | <b>Industry</b>      | <b>1 ADR=<br/>X shares</b> | <b>Date of<br/>ADR<br/>[primary]<br/>listing</b> |
|-------------------------------|--------------------------------|-----------------------------------------|----------------------|----------------------------|--------------------------------------------------|
| BSBR [SANB11]                 | Banco Santander S.A            | Brazil                                  | Fin Serv.            | 1                          | 10/09                                            |
| LND [AGRO3]*                  | BrasilAgro Companhia           | Brazil                                  | Food &<br>Drugs      | 1                          | 11/12                                            |
| OIBR [OIBR3]                  | Oi S.A                         | Brazil                                  |                      | 1                          | 11/01                                            |
| BAK [BRKM3]*                  | Braskem S.A                    | Brazil                                  |                      | 2                          | 12/98                                            |
| BRFS [BRFS3]*                 | BRF- Brasil Foods S.A          | Brazil                                  | Food &<br>Drugs      | 1                          | 10/00                                            |
| EBR [ELET3]                   | Centrais Electric- Electrobras | Brazil                                  | Power &<br>Utilities | 1                          | 10/08                                            |
| CBD [PCAR4]*                  | Companhia Brasileira           | Brazil                                  |                      | 1                          | 5/97                                             |
| ABV [AMBV3]*                  | AmBev                          | Brazil                                  |                      | 1                          | 6/97                                             |
| SBS [SBSP3]                   | Sabesp                         | Brazil                                  |                      | 1                          | 5/02                                             |
| CIG [CMIG3]                   | Cemig                          | Brazil                                  |                      | 1                          | 9/01                                             |
| ELP [CPLE3]*                  | COPEL                          | Brazil                                  |                      | 1                          | 7/97                                             |
| SID [CSNA3]                   | CSN                            | Brazil                                  |                      | 1                          | 11/97                                            |
| CPL [CPFE3]                   | CPFL                           | Brazil                                  |                      | 2                          | 9/04                                             |
| ERJ [EMBR3]                   | Embraer S.A                    | Brazil                                  |                      | 4                          | 7/00                                             |
| FBR [FIBR3]                   | Fibria Celulose S.A            | Brazil                                  |                      | 1                          | 4/00                                             |
| GFA [GFSA3]                   | Gafisa S.A                     | Brazil                                  |                      | 2                          | 3/07                                             |
| GGB [GGBR3]*                  | Gerdau S.A                     | Brazil                                  |                      | 1                          | 3/99                                             |
| GOL <sup>PRF</sup> GOLL4]*    | GOL                            | Brazil                                  |                      | 1                          | 6/04                                             |
| ITUB [ITUB3]*                 | Itau Unibanco S.A              | Brazil                                  |                      | 1                          | 2/02                                             |
| PBR [PETR3]                   | PETROBRAS                      | Brazil                                  | Oil & Gas            | 2                          | 8/00                                             |
| VIV [VIVT]                    | Telefonica Brasil S.A          | Brazil                                  | Telecom              | 1                          | 11/98                                            |
| TSU [TIMP3]                   | TIM S.A                        | Brazil                                  |                      | 5                          | 11/98                                            |
| UGP [UGPA3]                   | Ultrapar S.A                   | Brazil                                  |                      | 1                          | 10/99                                            |
| VALE [VALE3]                  | VALE S.A                       | Brazil                                  |                      | 1                          | 6/00                                             |
| PVD [PROVIDA]                 | A.F.P Provida S.A              | Chile                                   |                      | 15                         | 11/94                                            |
| BCH [CHILE]                   | Banco de Chile                 | Chile                                   | Fin Serv.            | 600                        | 1/02                                             |
| BSAC [BSAN]                   | Banco Santander-Chile          | Chile                                   | Fin Serv.            | 400                        | 1/97                                             |
| CNCO ()**                     | Cencosud SA                    | Chile                                   |                      |                            | 6/12                                             |
| CCU [CCU]                     | Compania .. Unidas S.A         | Chile                                   |                      | 2                          | 3/99                                             |
| BCA [CORPBANC]                | Corpbanca                      | Chile                                   | Fin Serv.            | 1500                       | 11/04                                            |
| AKO/A <sup>P</sup> [ANDINAA]* | Embotalladora Andina S.A       | Chile                                   |                      | 6                          | 7/94                                             |
| EOC [ENDESA]                  | Empresa Nacional de S.A        | Chile                                   |                      | 30                         | 7/94                                             |
| ENI [ENERSIS]                 | Enersis S.A                    | Chile                                   |                      | 50                         | 10/93<br>[3/13]                                  |
| LFL [LAN]                     | LAN Airlines S.A               | Chile                                   | Transport            | 1                          | 11/97                                            |
| SQM [SQM/B]                   | SQM                            | Chile                                   |                      | 1                          | 9/93                                             |

| Ticker                       | Name                                | Primary<br>Exchange<br>Country | Industry           | 1 ADR=<br>X shares | Date of<br>ADR<br>[primary]<br>listing |
|------------------------------|-------------------------------------|--------------------------------|--------------------|--------------------|----------------------------------------|
|                              |                                     |                                |                    |                    | [10/95]                                |
| VCO [CONCHA]                 | Vina Concha ...S.A                  | Chile                          |                    | 20                 | 10/94                                  |
| ACH [2600]                   | Aluminium Corp of China Ltd         | China (HK)                     | Metals &<br>Mining | 25                 | 12/01<br>[12/01]                       |
| CEA [0670]                   | China Eastern Airlines Corp Ltd     | China (HK)                     | Transport          | 50                 | 01/97<br>[02/97]                       |
| LFC [2628]                   | China Life Insurance Co Ltd         | China (HK)                     | Fin Serv           | 15                 | 12/03<br>[12/03]                       |
| CHA [0728]                   | China Telecom Corp Ltd              | China (HK)                     | Telecom            | 100                | 11/02<br>[11/02]                       |
| CHL [0941]**                 | China Mobile                        | China (HK)                     | Telecom            |                    |                                        |
| SNP [0386]                   | China Petroleum & Chem. Corp        | China (HK)                     | Oil & Gas          | 100                | 10/00<br>[10/00]                       |
| ZNH [1055]                   | China Southern Airlines Co Ltd      | China (HK)                     | Transport          | 50                 | 07/97<br>[07/97]                       |
| CEO [0883]**                 | CNOOC Ltd                           | China (HK)                     |                    | 100                | 02/01<br>[02/01]                       |
| GSH [0525]                   | Guangshen Railway Co Ltd            | China (HK)                     | Transport          | 50                 | 05/96<br>[05/96]                       |
| HNP [0902]**                 | Huaneng Power International Inc     | China (HK)                     | Energy             | 40                 | 08/03<br>[01/98]                       |
| PTR [0857]                   | PetroChina Co Ltd                   | China (HK)                     | Oil & Gas          | 100                | 03/00<br>[04/00]                       |
| SMI [981]                    | Semiconductor Mfg. Intl. Corp       | China (HK)                     |                    | 50                 | 03/04<br>[03/04]                       |
| SHI [0338]                   | Sinopec Shanghai Petroch. Co<br>Ltd | China (HK)                     | Oil & Gas          | 100                | 07/93<br>[07/93]                       |
| YZC [1171]                   | Yanzhou Coal Mining Co Ltd          | China (HK)                     | Metals &<br>Mining | 10                 | 03/98<br>[03/98]                       |
| CIB <sup>PR</sup> [PFBCOLO]* | BanColombia S.A                     | Colombia                       |                    | 4                  | 7/95                                   |
| EC*                          | Ecopetrol S.A                       | Colombia                       |                    | 20                 | 9/08<br>[11/07]                        |
| NBG []**                     | National Bank of Greece             | Greece                         | Fin Serv.          |                    | 10/99                                  |
| INFY [INFO]                  | Infosys Ltd                         | India                          |                    | 1                  | 12/12                                  |
| IBN [ICICIBC]                | ICICI Bank Ltd                      | India                          | Fin Serv.          | 2                  | 03/00                                  |
| TTM [TTMT]                   | Tata Motors Ltd                     | India                          | Mfg.               | 5                  | 09/04                                  |
| SSLT [STLT]**                | Sesa Sterlite Ltd                   | India                          |                    | 4                  | 06/07                                  |
| WIT [WPRO]                   | Wipro Ltd                           | India                          |                    | 1                  | 10/00                                  |
| HDB [HDFCB]                  | HDFC Bank Ltd                       | India                          | Fin. Serv.         | 3                  | 07/01                                  |

| <b>Ticker</b>    | <b>Name</b>                       | <b>Primary<br/>Exchange<br/>Country</b> | <b>Industry</b>    | <b>1 ADR=<br/>X shares</b> | <b>Date of<br/>ADR<br/>[primary]<br/>listing</b> |
|------------------|-----------------------------------|-----------------------------------------|--------------------|----------------------------|--------------------------------------------------|
| RDY [DRRD]       | Dr. Reddy's Laboratories Ltd      | India                                   | Food &<br>Drugs    | 1                          | 04/01                                            |
| WNS [ ]**        | WNS Holdings Ltd                  | India                                   |                    |                            | 07/06                                            |
| TLK [TLKM]       | Telekom Indonesia Pers Tbk PT     | Indonesia                               | Telecom            | 40                         | 11/95                                            |
| LPL [034220]     | LG Display Co. Ltd                | South Korea                             |                    | .5                         | 07/04                                            |
| SKM [017670]     | SK Telecom Co. Ltd                | South Korea                             | Telecom            | .11                        | 06/96                                            |
| KEP [015760]     | Korea Electric Power Co           | South Korea                             | Energy             | .5                         | 10/94                                            |
| PKX [005490]     | POSCO                             | South Korea                             |                    | .25                        | 10/94                                            |
| KT [030200]**    | KT Corp                           | South Korea                             |                    | .5                         | 05/99                                            |
| SHG [055550]     | Sinhan Financial Group Co. Ltd    | South Korea                             | Fin serv.          | 1                          | 09/03                                            |
| KB [105560]      | KB Financial Group Inc            | South Korea                             | Fin Serv.          | 1                          | 11/01                                            |
| WF [053000]      | Woori Finance Holdings Co. Ltd    | South Korea                             | Fin Serv.          | 3                          | 09/03                                            |
| AMX [AMXL]       | America Movil                     | Mexico                                  | Telecom            | 20                         | 2/01                                             |
| CX [CEMEXCPO]    | CEMEX                             | Mexico                                  |                    | 10                         | 9/99                                             |
| KOF [KOFL]*      | Coca Cola                         | Mexico                                  | Food &<br>Drugs    | 10                         | 9/93                                             |
| VLRS [ ]**       | Controladora ...                  |                                         |                    |                            | 9/13                                             |
| ICA [ICA]        | Empresas                          | Mexico                                  |                    | 4                          | 4/92 [8/05]                                      |
| FMX [FEMSAUBD]   | FEMSA                             | Mexico                                  |                    | 10                         | 5/98 [6/05]                                      |
| GMK [GRUMAB]     | Gruma                             | Mexico                                  |                    | 4                          | 11/98<br>10/96]                                  |
| PAC [GAPB]       | GAP                               | Mexico                                  |                    | 10                         | 2/06 [2/06]                                      |
| ASR [ASURB]      | ASUR                              | Mexico                                  |                    | 10                         | 9/00 [9/00]                                      |
| BSMX [SANMAXB]*  | Grupo Financiero                  | Mexico                                  |                    | 5                          | 9/12 [9/12]                                      |
| SIM [SIMECB]     | Grupo Simec                       | Mexico                                  |                    | 3                          | 7/93<br>[02/07]                                  |
| IBA [BACHOCOB]   | Industrias Bachoco                | Mexico                                  |                    | 12                         | 9/97                                             |
| MXT [MAXCOMCP]   | Maxcom Elecommunic.               | Mexico                                  | Telecom            | 7                          | 10/07<br>10/07]                                  |
| CPAC [CPACASC1]* | Cementos Pacasmayo S.A.A          | Peru                                    |                    | 5                          | 2/12 [7/99]                                      |
| BVN [BUENAVC1]*  | Compania de Minas Buenaven<br>S.A | Peru                                    |                    | 1                          | 5/96                                             |
| PHI [TELPM]      | Philippines Long Distance         | Philippines                             | Telecom            | 1                          | 10/94                                            |
| MTL [MTLR]*      | Mechel                            | Russia                                  |                    | 1                          | 10/04                                            |
| MBT [MTSS]*      | Mobile Telesystems                | Russia                                  | Telecom            | 2                          | 6/00<br>[03/09]                                  |
| AU [ANG]         | AngloGold Ashanti Limited         | South Africa                            | Metals &<br>Mining | 1                          | 8/98 [6/04]                                      |
| DRD [DRD]        | DRDGOLD Limited                   | South Africa                            | Metals &           | 10                         | 12/11                                            |

| Ticker       | Name                                    | Primary<br>Exchange<br>Country | Industry           | 1 ADR=<br>X shares | Date of<br>ADR<br>[primary]<br>listing |
|--------------|-----------------------------------------|--------------------------------|--------------------|--------------------|----------------------------------------|
|              |                                         |                                | Mining             |                    |                                        |
| GFI [GFI]    | Gold Fields Limited                     | South Africa                   | Metals &<br>Mining | 1                  | 5/93<br>[11/93]                        |
| HMY [HAR]    | Harmony Gold Mining Co<br>Limited       | South Africa                   | Metals &<br>Mining | 1                  | 05/95<br>[6/95]                        |
| SPP [SAP]    | Sappi Limited                           | South Africa                   |                    | 1                  | 09/95<br>[9/95]                        |
| SSL [SOL]    | Sasol Limited                           | South Africa                   |                    | 1                  | 06/93                                  |
| TSM [2330]   | Taiwan Semiconduct. Mfg. Co<br>Ltd      | Taiwan                         |                    | 5                  | 10/97                                  |
| HIMX [3222]* | Himax Technologies Inc                  | Taiwan                         |                    | 2                  |                                        |
| UMC [2303]   | United Microelectronics Corp            | Taiwan                         |                    | 5                  | 09/00                                  |
| AUO [2409]   | AU Optronics Corp                       | Taiwan                         |                    | 10                 | 05/02                                  |
| SPIL [2325]  | Siliconware Precision Industries.<br>Co | Taiwan                         |                    | 5                  | 09/00                                  |
| ASX [2311]   | Advanced Semiconductor Engn.<br>Inc     | Taiwan                         |                    | 5                  | 09/00                                  |
| CHT [2412]   | Chunghwa Telecom Co Ltd                 | Taiwan                         | Telecom            | 10                 | 07/03                                  |
| TKC [TCELL]* | Turkcell ...A.S                         | Turkey                         |                    | 2.5                | 7/00 [7/00]                            |
|              |                                         |                                |                    |                    |                                        |